Time-Varying Causal Survival Learning

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Abstract

Staggered adoption refers to settings in which individuals adopt an intervention at different times. Although many papers have studied staggered-adoption designs, little work has addressed survival outcomes. This paper fills that gap. We give causal assumptions under which a time-dependent survival model identifies a heterogeneous treatment effect. We then introduce a time-varying, event-risk propensity score. Constructing this propensity score is nontrivial because treatment is time-varying and the outcome is also a survival outcome, so the relevant propensity score must be defined relative to the contemporaneous risk set and event time. Existing propensity modifications do not generally account for this feature and therefore need not yield Neyman orthogonality for the Cox partial likelihood. We establish Neyman orthogonality and the corresponding double machine learning convergence rate. We also derive asymptotic linearity and an influence-function-based variance estimator for heterogeneous treatment effects in this setting. Simulations show that orthogonalization reduces estimation error and supports near-nominal inference when the baseline-risk model is complex and the sample size is sufficiently large. When baseline risk is linear, direct Cox estimation is more precise. We illustrate the framework using lawyer-level

Copilot adoption data, with an additional analysis of the Stanford Heart Transplant study in the Appendix.

1 Introduction

Many treatments and interventions are adopted at different times. A patient waits for an organ transplant, a customer starts a free trial, or a professional begins using a new software tool. Each unit therefore contributes untreated time before adoption and treated time afterward. When the outcome is itself a time to event, assigning everyone who eventually adopts to a baseline treated group incorrectly labels pre-adoption person-time as treated and can bias the treatment comparison.

The Stanford Heart Transplant study is the classical medical example (Crowley and Hu, 1977; Zhu et al., 2021). Patients entered the programme with severe heart disease and received a transplant only when a suitable donor became available. Waiting times ranged from days to more than a year (Evans et al., 1986; O'Connell et al., 1992). Similar timing arises outside medicine. In our main application, we study whether a lawyer's first use of an internally developed AI assistant changes the time to becoming a regular Copilot user. In both cases, there is no common date that separates treated and untreated units. We analyse the Copilot application in the main text and the Stanford Heart Transplant study in the Appendix.

Staggered-adoption designs compare treated units with units that have not yet adopted (Athey and Imbens, 2022). Existing designs are largely developed for repeatedly measured outcomes, such as annual GDP, for which the same unit contributes several pre- and post-adoption observations. A survival outcome is different: the event time is observed once, or is censored. We connect the two settings by expressing the staggered-adoption restrictions

in terms of counterfactual hazards. No anticipation makes the pre-adoption hazard equal to the never-treated hazard, and invariance to treatment history makes the post-adoption hazard depend on current treatment status rather than the exact adoption date. Together with standard causal assumptions and the proportional-hazards restriction, these conditions give a causal interpretation to a conditional log hazard ratio. This interpretation requires care because hazard ratios compare subjects who remain at risk over time (Hernán, 2010; Aalen et al., 2015; Martinussen et al., 2020), as discussed in Section 2.2.

Flexible estimation of baseline risk creates a second problem. In a direct time-dependent Cox fit, estimation error from a regularized or flexible model for baseline risk can enter the treatment-effect score at first order. We therefore use double machine learning (DML) to construct a partial-likelihood score whose first derivative with respect to the nuisance functions is zero (Chernozhukov et al., 2018). The key is to find the correct propensity score for treatment centering. We show that this is the probability of having adopted treatment among subjects who experience the event at time t, conditional on their covariates. This differs from the usual adoption propensity because it also accounts for survival selection and the relative event hazards. Centering by this event propensity makes the Cox partial-likelihood score Neyman orthogonal.

The paper makes three contributions. First, it formulates a staggered-adoption design for time-to-event outcomes and states the assumptions under which the heterogeneous log hazard ratio has a causal interpretation. Second, and most importantly, it derives the event-risk propensity that makes the Cox partial-likelihood score Neyman orthogonal. The resulting estimator has a second-order nuisance remainder and a DML convergence rate for a finite-dimensional heterogeneous effect. The event-time-pooled modification of Gao and Hastie (2025) does not generally satisfy the required risk-set centering identity and therefore need not orthogonalize the Cox partial-likelihood score. Third, we establish

asymptotic linearity and provide conditions for valid sandwich variance estimation and pointwise confidence intervals for the heterogeneous log hazard ratio.

Our simulations show when orthogonalization is useful. When baseline log risk is linear, an ordinary time-dependent Cox regression is simpler and more precise. At $n = 2000$, its MSE is 0.018, compared with 0.030 for TV-CSL with the time-varying propensity. When baseline risk is non-linear and fitted flexibly, TV-CSL performs better at sufficiently large sample sizes. At $n = 2000$, its MSE is 0.042, compared with 0.094 for Cox, and its empirical 95% coverage is 0.933, compared with 0.545 for Cox. An additional simulation shows that the event-time propensity reduces bias and improves coverage relative to marginal centering as the treatment effect increases.

In the Copilot application, we find that first use of the firm's AI assistant is associated with a higher rate of becoming a regular Copilot user. Using TV-CSL with the time-varying propensity, the estimated hazard ratio is 1.51 (95% CI: 1.28–1.79) for senior lawyers and 1.23 (1.02–1.48) for junior lawyers. These findings suggest a positive association between adoption of one AI tool and subsequent regular use of another. The causal interpretation of these estimates relies on the proportional-hazards and event-time overlap assumptions, whose limitations we discuss below.

1.1 Literature Review

Our work builds on survival analysis, staggered-adoption designs, and heterogeneous treatment-effect estimation. Survival analysis provides tools for time-to-event data, while recent causal work studies treatments that change over time. Their intersection, causal inference for survival outcomes with a one-time treatment adopted at a stochastic time, is less developed.

Several strands of causal inference address survival outcomes. Robins (1992) studies time-dependent accelerated failure-time models in the presence of confounding, while Ab-

bring and van den Berg (2003) study identification of treatment effects in duration models. Hernán (2010), Aalen et al. (2015), and Martinussen et al. (2020) discuss the causal interpretation of hazard ratios, and Vansteelandt et al. (2024) develop assumption-lean Cox regression. Gran et al. (2010) and Keogh et al. (2023) study sequential Cox models and trial-based approaches for handling time-dependent confounding. Our treatment instead changes once, at a stochastic time, and then remains on. Each failure therefore compares already-treated and not-yet-treated subjects in the contemporaneous risk set.

Treatments with random timing have been studied extensively in the difference-in-differences literature. Athey and Imbens (2022) and Goodman-Bacon (2021) emphasize the role of treatment timing in panel designs, while Sun and Abraham (2021) studies bias from effect heterogeneity across adoption cohorts and Imai and Kim (2021) develops related methods for panel data. These methods do not directly address a once-observed, possibly censored event time.

For Cox regression, Andersen and Gill (1982) establish the large-sample theory using counting processes, and Fisher and Lin (1999) discuss time-dependent covariates. Fang et al. (2017) develop inference using decorrelated scores for high-dimensional Cox models. Other causal uses of the Cox model answer different questions. Shaikh and Toulis (2021) develop randomization tests for staggered adoption, whereas we estimate effect heterogeneity. Zhu and Gallego (2022) consider electronic-health-record settings with longitudinal covariates; our identifying assumptions instead concern one-time adoption and baseline confounding.

For heterogeneous treatment-effect estimation with time-to-event outcomes, Cui et al. (2023) develop causal survival forests using doubly robust estimating equations, and Xu et al. (2024) propose censoring-unbiased transformations that can be combined with existing HTE learners. Xu et al. (2022) review extensions of metalearners to right-censored

outcomes. Gao and Hastie (2025) develop a DML construction for exponential families and discuss the Cox model as a special case. Unlike causal survival forests formulated for baseline treatment and the event-pooled propensity modification of Gao and Hastie (2025), we study treatment adopted at a stochastic time and derive the event-time propensity required for Neyman orthogonality of the Cox partial likelihood.

Section 2 introduces the causal model and target. Section 3 reviews the time-dependent Cox partial likelihood, and Section 4 develops TV-CSL and its inference theory. Section 5 reports the simulations, Section 6 describes the Copilot adoption application, and Section 7 concludes. The event-propensity derivation, proofs, complete simulation tables, and Stanford Heart Transplant analysis are in the Appendix.

2 Problem Setup

2.1 Notation and observed data

Let capital letters denote random variables and lowercase letters their realizations. For subject i, let $T_i(a) \in [0, \infty]$ be the potential event time if treatment is first adopted at time $a \in [0, \infty]$; $a = \infty$ denotes no adoption during follow-up. We observe the adoption time A_i and baseline covariates $X_i \in \mathbb{R}^p$. The associated predictable treatment process is

$$W_i(t) = \mathbf{1}(A_i < t).$$

The strict inequality records treatment status immediately before t and is immaterial when time is continuous and ties have probability zero.

Let C_i be the censoring time. We observe

$$O_i = (X_i, A_i, U_i, \Delta_i), \qquad U_i = T_i \wedge C_i, \qquad \Delta_i = \mathbf{1}(T_i \leq C_i).$$

Censoring can arise from loss to follow-up or administrative study end; for a censored subject, we know only that $T_i > C_i$. We define the at-risk and event counting processes as

$$Y_i(t) = \mathbf{1}(t \leq U_i), \qquad N_i(t) = \mathbf{1}(U_i \leq t, \Delta_i = 1).$$

We adapt three standard causal assumptions (Rubin, 1974).

Assumption 1 (SUTVA)**.** There is no interference between subjects and treatment has a well-defined version. The observed event time satisfies $T_i = T_i(A_i)$.

Assumption 2 (Unconfounded adoption time)**.** For every a,

$$A_i \perp\!\!\!\perp T_i(a) \mid X_i.$$

Thus the measured baseline covariates are sufficient to compare subjects with different adoption times.

Assumption 3 (Overlap)**.** There exist $\epsilon > 0$ and $0 < t_0 < \bar{t} < \infty$ such that, for almost every x and all $t \in [t_0, \bar{t}]$,

$$\epsilon \leq \pi_t(x) := P(A < t \mid X = x) \leq 1 - \epsilon,$$

and

$$P(U \geq \bar{t} \mid X = x, A) \geq \epsilon$$

for almost every adoption time A.

The first requirement ensures variation in treatment adoption. The second ensures that both treatment groups can remain represented in the risk sets throughout the analysis horizon; without it, survival or censoring could eliminate treatment possibility of one arm from the comparisons underlying the Cox partial likelihood.

2.2 Hazard restrictions and causal target

Without parametric restrictions, write

$$X_i \sim f_X, \qquad A_i \mid X_i \sim k(\cdot \mid X_i), \qquad T_i(a) \mid X_i \sim f(\cdot \mid a, X_i).$$

For a fixed adoption time a and covariate value x, the counterfactual hazard is

$$h(t \mid a, x) = \lim_{\delta \downarrow 0} \frac{P\{t \le T_i(a) < t + \delta \mid T_i(a) \ge t, X_i = x\}}{\delta}.$$

The hazard and event-time distribution uniquely determine one another. For a generic hazard $h(t)$, let $\Lambda(t) = \int_0^t h(s)\, ds$ denote the cumulative hazard. Then

$$1 - F(t) = e^{-\Lambda(t)}, \qquad f(t) = h(t)e^{-\Lambda(t)},$$

and conversely $h(t) = f(t)/\{1 - F(t)\}$ (Klein and Moeschberger, 1997).

There is one counterfactual hazard for every possible adoption time. Two restrictions reduce this collection to hazards under current treatment status.

Assumption 4 (No anticipation)**.** For $t < a$,

$$h(t \mid a, x) = h(t \mid \infty, x).$$

No anticipation means before adoption, a future adoption date does not change the current hazard (Abbring, 2008).

Assumption 5 (Invariance to adoption history)**.** For $t > a$,

$$h(t \mid a, x) = h(t \mid 0, x).$$

This assumption means after adoption, the hazard depends on current treatment status but not on the exact adoption date or time since adoption. Related restrictions are used in staggered-adoption designs; see Section 3.2 of Athey and Imbens (2022).

Define the untreated and treated hazards by

$$h_0(t \mid x) = h(t \mid \infty, x), \qquad h_1(t \mid x) = h(t \mid 0, x).$$

Under Assumptions 4 and 5,

$$h(t \mid a, x) = h_0(t \mid x) \left\{ \frac{h_1(t \mid x)}{h_0(t \mid x)} \right\}^{\mathbf{1}(a<t)}.$$

We impose the proportional-hazards model (Cox, 1972)

$$\begin{aligned} h_0(t \mid x) &= \lambda_0(t) e^{\eta_0(x)}, \\ h_1(t \mid x) &= \lambda_0(t) e^{\eta_1(x)}. \end{aligned} \tag{1}$$

Here $\lambda_0(t)$ is the unspecified baseline hazard, $\Lambda_0(t) = \int_0^t \lambda_0(s)\, ds$ is its cumulative hazard, and $\eta_w(x)$ is the covariate-dependent log relative risk under treatment status w. The baseline hazard cancels from the treatment contrast The time component cancels from the contrast

$$\tau_0(x) = \log\left\{\frac{h_1(t \mid x)}{h_0(t \mid x)}\right\} = \eta_1(x) - \eta_0(x). \tag{2}$$

We call $\tau_0(x)$ the heterogeneous log hazard ratio (HLHR), following the use of covariate-specific hazard-ratio targets in survival HTE estimation (Kalbfleisch and Prentice, 2011; Gao and Hastie, 2025).

We assume that the HLHR is linear in a fixed, user-chosen basis $p(x) \in \mathbb{R}^d$:

$$\tau_0(x) = p(x)^\top \beta_0. \tag{3}$$

The basis may contain original covariates, transformations, or prespecified interactions. This restriction keeps the target finite-dimensional while allowing $\eta_0(x)$ to be learned flexibly. The causal interpretation of a hazard ratio requires care because conditioning on survival can induce selection over time (Hernán, 2010). Under the proportional-hazards model above, this selection is incorporated into the model through the covariate-specific, time-invariant contrast $\tau_0(x)$; when proportional hazards fails, however, $\tau_0(x)$ should instead be viewed as a working summary rather than a time-invariant causal effect.

The observed time-varying hazard can now be written as

$$h(t \mid A_i = a, X_i = x) = \lambda_0(t) \exp\{\eta_0(x) + \mathbf{1}(a < t) p(x)^\top \beta_0\}. \tag{4}$$

Our parameter of interest is β. η is the nuisance parameter. Section 3 first reviews estimation procedure of β under the time-varying hazard when η is simple. Section 4 then

develops a valid inference procedure for β_0 when η_0 has a complex structure where an orthogonal score equation is derived for double machine learning.

3 Time-Dependent Cox Partial Likelihood

We review how staggered adoption changes the Cox partial likelihood by comparing fixed and time-varying treatment (Fisher and Lin, 1999; Kalbfleisch and Prentice, 2011).

With treatment fixed at baseline, consider

$$h(t \mid w, x) = \lambda_0(t) \exp\{\eta_0(x) + w\tau_0(x)\}, \tag{5}$$

with the same treatment status used throughout follow-up. At an observed event time U_i, define the risk set

$$\mathcal{R}_i = \{j : Y_j(U_i) = 1\} = \{j : U_j \geq U_i\}, \tag{6}$$

consisting of subjects who remain event-free and uncensored immediately before U_i.

Conditional on one subject in $\mathcal{R}_i$ failing (meaning an event is observed) at U_i, the partial-likelihood contribution is the hazard of subject i divided by the sum of the hazards over $\mathcal{R}_i$. The log of the product of these contributions is (Cox, 1972; Andersen et al., 1993)

$$\begin{aligned} \mathrm{pl}_n(\tau, \eta_0) = \frac{1}{n} \sum_{\Delta_i = 1} \Bigg[& \eta_0(X_i) + W_i \tau(X_i) \\ & - \log \left\{ \sum_{j \in \mathcal{R}_i} \exp\big(\eta_0(X_j) + W_j \tau(X_j)\big) \right\} \Bigg]. \end{aligned} \tag{7}$$

For staggered adoption, treatment status must instead be read at each event time. The key distinction is the treatment indicator: the fixed model uses the same W_j at every event

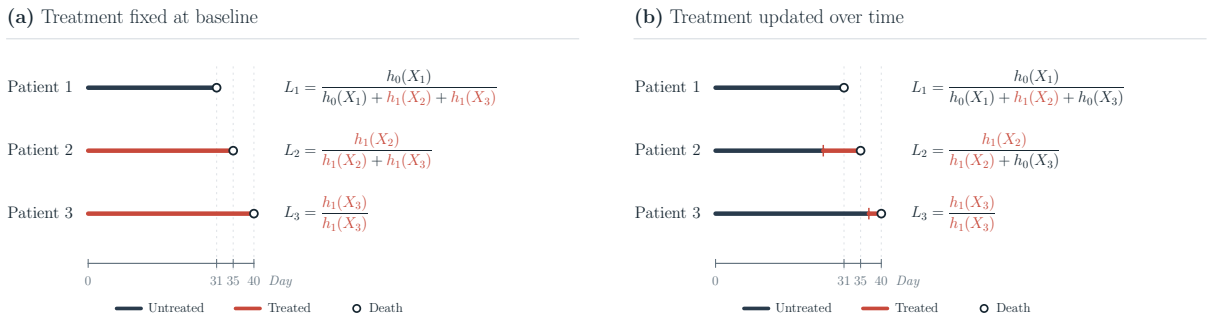


Figure 1: Partial-likelihood contributions with fixed and time-varying treatment. Black denotes untreated person-time and red denotes treated person-time. Panel (a) treats eventual adopters as treated from baseline, which shows that the control group appears to have shorter observed survival times on average. In contrast, panel (b) updates treatment status at the actual adoption times, which shows that treated time before death is shorter than untreated time on average. This contrast illustrates how ignoring treatment timing can reverse the treatment effect conclusion and why each Cox partial-likelihood denominator must use the treatment status of every subject at the corresponding event time.

time, whereas staggered adoption requires the time-updated $W_j(U_i)$ for each subject in the risk set. The resulting partial likelihood is

$$\mathrm{pl}_n(\tau, \eta_0) = \frac{1}{n} \sum_{\Delta_i=1} \Bigg[\eta_0(X_i) + W_i(U_i)\tau(X_i) - \log \Bigg\{ \sum_{j \in \mathcal{R}_i} \exp\big(\eta_0(X_j) + W_j(U_i)\tau(X_j)\big) \Bigg\} \Bigg]. \tag{8}$$

In Equation 7, each subject's contribution is fixed at either $\eta_0(X_j)$ or $\eta_1(X_j)$ across all event times. In contrast, Equation 8 allows the same subject's contribution to change from $\eta_0(X_j)$ before adoption to $\eta_1(X_j)$ afterward. Figure 1 illustrates the resulting change in the risk-set denominators.

We impose the following conditions for consistency of the time-dependent Cox partial-

likelihood estimator.

We impose the following standard regularity conditions for time-dependent Cox regression.

Condition 1 (Cox regularity)**.** (i) The observations are i.i.d. by subject. Treatment status $W_i(t)$ is determined immediately before t, and the observed event process has conditional intensity

$$Y_i(t)e^{\eta_{W_i(t)}(X_i)}\, d\Lambda_0(t).$$

(ii) The cumulative baseline hazard $\Lambda_0(\bar{t})$ is finite.

(iii) The population hazard-weighted risk-set denominator

$$c_t := E[Y(t)e^{\eta_{W(t)}(X)}]$$

satisfies $\inf_{t\in[0,\bar{t}]} c_t > 0$.

Condition (i) includes the required non-informative censoring assumption and the use of treatment information available before each event time. Conditions (ii) and (iii) ensure finite follow-up intensity and a nonvanishing weighted risk set (Andersen et al., 1993; Fleming and Harrington, 2005).

Proposition 1 (Consistency of the time-dependent Cox partial likelihood)**.** *Under Assumptions 1–5 and Condition 1, suppose $\eta_0(x) = b(x)^\top\gamma_0$ is correctly specified with a fixed-dimensional basis. If the partial-likelihood information for (γ_0, β_0) is nonsingular and the standard Cox moment conditions hold, then any regular maximizer of Equation 8 satisfies*

$$(\hat{\gamma}, \hat{\beta}) \xrightarrow{p} (\gamma_0, \beta_0).$$

The proposition is the standard consistency result for a Cox model with predictable time-dependent covariates. It follows from the martingale representation of the score and a uniform law of large numbers for the normalized partial likelihood (Andersen et al., 1993; Fleming and Harrington, 2005).

The direct estimator is attractive when a low-dimensional model for η_0 is adequate. When baseline risk is learned by a slower, flexible procedure, however, first-stage error can enter estimation of β_0 at first order. The next section changes the parametrization so that this first-order effect cancels.

4 Double Machine Learning Estimator

Equation (8) estimates baseline risk and treatment-effect heterogeneity jointly. TV-CSL instead allows the baseline-risk function to be estimated flexibly while protecting the finite-dimensional treatment-effect estimator against first-order nuisance estimation error (Chernozhukov et al., 2018; Gao and Hastie, 2025).

We first derive the risk-set centering and its event-propensity interpretation. We then present the cross-fitted estimator and establish Neyman orthogonality and its convergence rate. Finally, we develop inference for the heterogeneous effect.

4.1 Risk-set centering and event propensity

Recall that $\eta_1(x) - \eta_0(x) = p(x)^\top \beta_0$. For any measurable function $a(t, x)$, define

$$\nu_a(t, x) = a(t, x)\eta_1(x) + \{1 - a(t, x)\}\eta_0(x).$$

Then

$$\eta_w(x) = \nu_a(t,x) + \{w - a(t,x)\}p(x)^\top \beta_0, \qquad w \in \{0,1\}.$$

Although every choice of a gives an exact reparametrization, we seek one that makes treatment-effect estimation insensitive to first-order nuisance errors. As established below, the required centering identity is

$$E\left[Y(t)e^{\eta_{W(t)}(X)}\{W(t) - a_0(t,X)\}p(X) \mid X\right] = 0. \tag{9}$$

To solve this equation, define

$$R_w(t,x) = P\{Y(t) = 1, W(t) = w \mid X = x\}, \qquad w \in \{0,1\}.$$

These are the treatment-specific risk-set probabilities: the proportions of individuals with covariates x who remain at risk at t in each treatment state.

The unique scalar centering solution on the event-risk support is

$$\begin{aligned} a_0(t,x) &= \frac{R_1(t,x)e^{\eta_1(x)}}{R_1(t,x)e^{\eta_1(x)} + R_0(t,x)e^{\eta_0(x)}}, \\ \nu_0(t,x) &= \eta_0(x) + a_0(t,x)\tau_0(x). \end{aligned} \tag{10}$$

To the best of our knowledge, $a_0(t,x)$ is a new type of propensity score in the literature. To aid interpretation, we now provide two ways to understand it.

A hazard-weighted change of measure. The centering has a natural interpretation as a change of measure. Using c_t from Condition 1, define

$$dP_{R_t}(O) = \frac{Y(t)e^{\eta_{W(t)}(X)}}{c_t}\, dP(O). \tag{11}$$

Write E_{R_t} for expectation under P_{R_t}. Equation (9) is equivalent to

$$E_{R_t}\left[\{W(t) - a_0(t, X)\}p(X) \mid X\right] = 0,$$

and therefore

$$a_0(t, x) = E_{R_t}[W(t) \mid X = x]. \tag{12}$$

The measure P_{R_t} first restricts attention to subjects at risk at time t and then weights them by their relative event hazards. Consequently, $a_0(t, x)$ describes treatment among subjects contributing to events at time t, rather than treatment among all subjects or merely among those still at risk. This is the distribution underlying the Cox partial-likelihood comparison.

Observable event propensity. By the counting-process intensity and Bayes' rule, Equation (12) also admits the characterization

$$a_0(t, x) = P\{W(t) = 1 \mid dN(t) = 1, X = x\}. \tag{13}$$

The conditioning is understood as a limiting probability over an infinitesimal event-time interval; its derivation is given in Appendix A.

This representation is useful for estimation: among observed failures, we can regress treatment status $W_i(U_i)$, a binary variable, on (U_i, X_i), without separately estimating R_0, R_1, or the treatment-specific hazards. In our simulation and data analysis, we fit a logistic generalized additive model.

Although $a_0(t, x)$ differs from the ordinary adoption propensity $\pi_t(x)$, the overlap assumption also controls event-time overlap.

Lemma 4.1 (Event-time overlap)**.** *Under Equation* (4)*,*

$$\operatorname{logit} a_0(t,x) = \operatorname{logit} \pi_t(x) + \tau_0(x) + \log \rho_t(x),$$

where

$$\rho_t(x) = \frac{P(U \geq t \mid A < t, X = x)}{P(U \geq t \mid A \geq t, X = x)}.$$

Under Assumption 3 and $\|\tau_0\|_\infty < \infty$*,* $a_0(t,x)$ *is bounded away from zero and one on* $[t_0, \bar{t}]$*.*

Our centering differs from the modified propensity of Gao and Hastie (2025). For baseline treatment W, their construction uses

$$a^{\mathrm{GH}}(x) = P(W = 1 \mid \Delta = 1, X = x). \tag{14}$$

Its direct staggered-adoption analogue is

$$a_0^{\mathrm{GH}}(t,x) = P(A < t \mid \Delta = 1, X = x). \tag{15}$$

This conditions on an observed failure at any time during follow-up, rather than a failure at the particular time t. It therefore need not satisfy Equation (9) or yield a Neyman-orthogonal Cox partial-likelihood score, as pointed out by the authors. In the simulations, we show that $a^{\mathrm{GH}}(x)$ can work well in some settings but not in others, whereas $a_0(t,x)$ is the centering implied by the orthogonality theory.

Figure 2 illustrates the nested populations underlying the three propensity definitions: the full population with $X = x$, the subset experiencing an observed failure ($\Delta = 1$), and the smaller subset experiencing a failure at time t ($dN(t) = 1$).

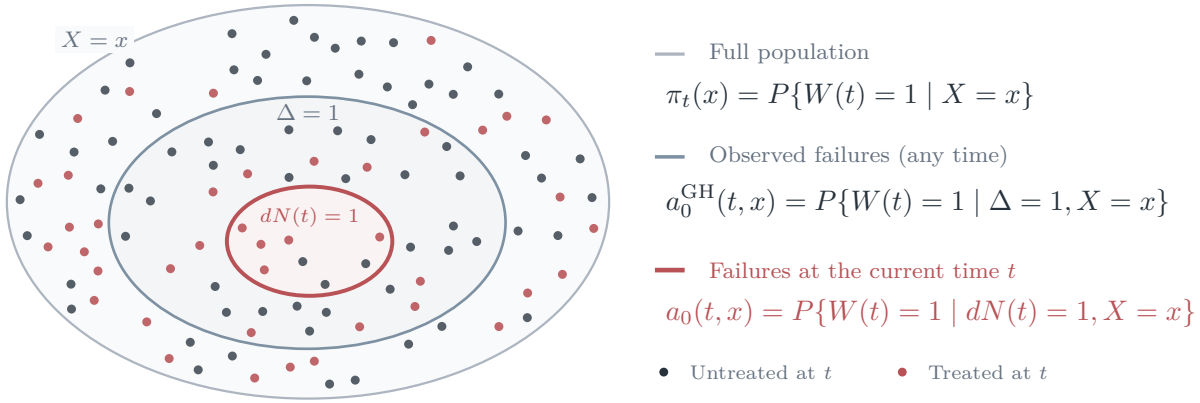


Figure 2: Nested populations underlying three propensity definitions at a fixed time t and covariate value $X = x$. The ordinary propensity $\pi_t(x)$ describes treatment status in the full population; $a_0^{\mathrm{GH}}(t, x)$ restricts attention to observed failures across follow-up; and $a_0(t, x)$ further conditions on failure at time t. The latter is understood through an infinitesimal event-time interval and provides the required Neyman orthogonality for Cox partial-likelihood. Red and dark dots denote treated and untreated individuals, respectively. Population sizes and treatment proportions are schematic.

4.2 Time-Varying Causal Survival Learner (TV-CSL)

We now construct the cross-fitted estimator and establish its theoretical properties.

For a candidate centering function a, define

$$q_a(t) = \{W(t) - a(t, X)\}p(X), \qquad q_0(t) = q_{a_0}(t).$$

Let

$$\bar{q}(t; \beta, a, \nu) = \frac{E[Y(t)e^{\nu(t,X)+q_a(t)^\top \beta} q_a(t)]}{E[Y(t)e^{\nu(t,X)+q_a(t)^\top \beta}]}.$$

The population partial-likelihood score is

$$\Psi(\beta, a, \nu) = E \int_0^{\bar{t}} \{q_a(t) - \bar{q}(t; \beta, a, \nu)\}\, dN(t). \tag{16}$$

Cross-fitted estimation. Partition the subjects into K folds $I_1, \ldots, I_K$, with $n_k = |I_k|$. For each fold k, estimate $\hat{a}^{(-k)}$ and $\hat{\nu}^{(-k)}$ using only I_k^c. For $i \in I_k$, set

$$\begin{aligned} \hat{q}_i(t) &= \{W_i(t) - \hat{a}^{(-k)}(t, X_i)\} p(X_i), \\ \hat{\theta}_i(t; \beta) &= \hat{\nu}^{(-k)}(t, X_i) + \hat{q}_i(t)^\top \beta. \end{aligned} \tag{17}$$

Define the held-out risk-set sums

$$\begin{aligned} \hat{S}_k^{(m)}(t; \beta) &= \frac{1}{n_k} \sum_{j \in I_k} Y_j(t) e^{\hat{\theta}_j(t;\beta)} \hat{q}_j(t)^{\otimes m}, \quad m = 0, 1, 2, \\ \hat{\bar{q}}_k(t; \beta) &= \frac{\hat{S}_k^{(1)}(t; \beta)}{\hat{S}_k^{(0)}(t; \beta)}, \end{aligned} \tag{18}$$

where $q^{\otimes 0} = 1$, $q^{\otimes 1} = q$, and $q^{\otimes 2} = qq^\top$.

The estimator solves

$$\begin{aligned} 0 = \hat{\Psi}_n(\hat{\beta}) &= \sum_{k=1}^{K} \frac{n_k}{n} \hat{\Psi}_k(\hat{\beta}), \\ \hat{\Psi}_k(\beta) &= \frac{1}{n_k} \sum_{i \in I_k} \int_0^{\bar{t}} \{\hat{q}_i(t) - \hat{\bar{q}}_k(t; \beta)\}\, dN_i(t). \end{aligned} \tag{19}$$

Algorithm 1 summarizes the estimation procedure. The nuisance $\hat{\nu}$ is held fixed as an offset when solving the second-stage Cox score.

If $\hat{\eta}_1$ is obtained as $\hat{\eta}_0 + p^\top \tilde{\beta}$, the preliminary estimate $\tilde{\beta}$ is fitted within the corresponding

Algorithm 1 Risk-set-orthogonalized TV-CSL

Input: $\{O_i\}_{i=1}^n$, basis $p(\cdot)$, subject-level folds $I_1, \ldots, I_K$, and horizon $\bar{t}$.
for $k = 1, \ldots, K$ **do**
    On I_k^c, estimate $a_0(t, x)$ by binary regression of $W_i(U_i)$ on (U_i, X_i) among observed failures, using Equation (13).
    On I_k^c, estimate η_0, η_1 using a preliminary time-dependent survival fit. Construct $\hat{\nu}^{(-k)}$ using Equation (10) with the fitted functions.
    Construct the held-out regressors and risk-set sums using Equations (17) and (18).
end for
Solve Equation (19) for $\hat{\beta}$.
Output: $\hat{\beta}$ and $\hat{\tau}(x) = p(x)^\top \hat{\beta}$.

training fold.

Theoretical properties. The treatment information matrix is

$$J = \int_0^{\bar{t}} c_t E_{R_t}[q_0(t) q_0(t)^\top] \, d\Lambda_0(t). \tag{20}$$

We impose Condition 1 and the following additional conditions. The event-weighted norm $\|\cdot\|_{2,\mu}$ is defined in Appendix B.1.

Condition 2 (DML regularity)**.** (i) $E\|p(X)\|^4 < \infty$, and the eigenvalues of J are bounded away from zero and infinity.

(ii) The cross-fitted nuisance estimates satisfy

$$\|\hat{a} - a_0\|_{2,\mu} = O_p(r_{a,n}), \qquad \|\hat{\nu} - \nu_0\|_{2,\mu} = O_p(r_{\nu,n}), \tag{21}$$

where $r_{a,n} + r_{\nu,n} = o(1)$.

Additional technical conditions are stated in Appendix B.1. Cross-fitting permits flex-

ible nuisance learners without requiring Donsker conditions on their function classes.

Proposition 2 (Neyman orthogonality)**.** *Suppose Conditions 1 and 3(ii) hold. At the true parameters,*

$$\Psi(\beta_0, a_0, \nu) = 0$$

for every admissible ν. Moreover, for all admissible directions h_a, h_ν,

$$D_a\Psi(\beta_0, a_0, \nu_0)[h_a] = 0, \qquad D_\nu\Psi(\beta_0, a_0, \nu_0)[h_\nu] = 0.$$

The proof is given in Appendix B.2. The result is stronger than ordinary first-order insensitivity to ν: when $a = a_0$, the population score vanishes for every admissible ν. This removes the pure $r_{\nu,n}^2$ term from the nuisance remainder.

Proposition 3 (DML convergence rate)**.** *Suppose Conditions 1, 2, and 3 hold. Then*

$$\|\hat{\beta} - \beta_0\| = O_p\left\{n^{-1/2} + r_{a,n}(r_{a,n} + r_{\nu,n})\right\}.$$

In particular, root-n consistency follows if

$$\sqrt{n}\, r_{a,n}(r_{a,n} + r_{\nu,n}) \to 0.$$

A sufficient condition is $r_{a,n} = o(n^{-1/4})$ and $r_{\nu,n} = o(n^{-1/4})$.

The proof is in Appendix B.3. The rate is asymmetric: accurate event-time centering allows slower convergence of the outcome nuisance, but not conversely.

4.3 Inference with TV-CSL

The influence-function representation uses the event martingale

$$M_i(t) = N_i(t) - \int_0^t Y_i(s) e^{\eta_{W_i(s)}(X_i)}\, d\Lambda_0(s).$$

Writing $q_i^0(t)$ for the subject-specific realization of $q_0(t)$, define

$$\psi_i = \int_0^{\bar{t}} q_i^0(t)\, dM_i(t), \qquad \Sigma = E(\psi_i \psi_i^\top), \qquad \Omega = J^{-1}\Sigma J^{-1}.$$

Here J is defined in Equation (20). The influence contribution ψ_i differs from the population score in Equation (16): it integrates the centered regressor against the event martingale rather than the observed counting process.

Proposition 4 (Asymptotic normality)**.** *Suppose the conditions of Proposition 3 hold and*

$$\sqrt{n}\, r_{a,n}(r_{a,n} + r_{\nu,n}) \to 0.$$

Then

$$\sqrt{n}(\hat{\beta} - \beta_0) = J^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi_i + o_p(1) \;\rightsquigarrow\; N(0, \Omega).$$

The proof is given in Appendix B.4. This is the Cox partial-likelihood analogue of best-linear-predictor inference in Semenova and Chernozhukov (2021).

Variance estimation. Using the risk-set sums in Equation (18), define

$$\hat{V}_k(t) = \frac{\hat{S}_k^{(2)}(t;\hat{\beta})}{\hat{S}_k^{(0)}(t;\hat{\beta})} - \hat{\bar{q}}_k(t;\hat{\beta})\hat{\bar{q}}_k(t;\hat{\beta})^\top,$$

and

$$\hat{J} = \frac{1}{n} \sum_{k=1}^{K} \sum_{\substack{i \in I_k \\ \Delta_i = 1}} \hat{V}_k(U_i).$$

For $i \in I_k$, the fold-specific Breslow increment and martingale residual are

$$d\hat{\Lambda}_{0,k}(t) = \frac{\sum_{i \in I_k} dN_i(t)}{\sum_{j \in I_k} Y_j(t) e^{\hat{\theta}_j(t;\hat{\beta})}},$$

$$d\hat{M}_i(t) = dN_i(t) - Y_i(t) e^{\hat{\theta}_i(t;\hat{\beta})} \, d\hat{\Lambda}_{0,k}(t).$$

Define

$$\hat{\psi}_i = \int_0^{\bar{t}} \{\hat{q}_i(t) - \hat{\bar{q}}_k(t;\hat{\beta})\} \, d\hat{M}_i(t),$$

and the covariance estimators

$$\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^{n} \hat{\psi}_i \hat{\psi}_i^\top, \qquad \hat{\Omega} = \hat{J}^{-1} \hat{\Sigma} \hat{J}^{-1}.$$

The sandwich standard error for $\hat{\beta}_j$ is $\{\hat{\Omega}_{jj}/n\}^{1/2}$. Under the correctly specified Cox intensity, $\Sigma = J$, giving the information-based alternative $\{\hat{J}_{jj}^{-1}/n\}^{1/2}$. We report both in the simulations.

For any covariate value x, pointwise inference for the heterogeneous log hazard ratio uses

$$\widehat{\mathrm{se}}\{\hat{\tau}(x)\} = \left\{ \frac{p(x)^\top \hat{\Omega} p(x)}{n} \right\}^{1/2},$$

giving the $(1-\alpha)$ confidence interval

$$p(x)^\top \hat{\beta} \;\pm\; z_{1-\alpha/2} \left\{ \frac{p(x)^\top \hat{\Omega} p(x)}{n} \right\}^{1/2}.$$

5 Simulation Study

In the simulation study, we investigate when TV-CSL improves on simpler time-dependent Cox estimators and when it does not. We have two conclusions. First, for point estimation, TV-CSL achieves lower mean squared error when the baseline risk function η_0 is complex and the sample size is sufficiently large. When baseline risk is simple, or when the sample size is small, direct Cox estimation is good enough. Second, for inference, TV-CSL orthogonalization is almost always necessary to achieve reliable coverage. The rest of the section provides details.

5.1 Simulation Design

For $i = 1, \ldots, n$, let $X_i = (X_{i1}, X_{i2}, X_{i3})^\top$ draw from independent Normal$(0, 1)$, and generate the adoption time from

$$A_i \mid X_i = x \sim \text{Exp}\{r(x)\}, \qquad r(x) = e^{x_2 + x_3}.$$

The event hazard follows

$$h(t \mid a, x) = t \exp\{\eta_0(x) + \mathbf{1}(a < t)\tau_0(x)\}, \qquad \tau_0(x) = p(x)^\top \beta_0 = x_1 + x_2 + x_3,$$

where $p(x) = (1, x_1, x_2, x_3)^\top$ and $\beta_0 = (0, 1, 1, 1)^\top$. We consider two baseline log-hazard functions. The first is linear,

$$\eta_0^L(x) = 2.5(x_1 + x_2/2 + x_3/3),$$

The second is non-linear,

$$\eta_0^{NL}(x) = -\tfrac{3}{4}\varsigma(x_1)\varsigma(x_2), \qquad \varsigma(z) = \frac{2}{1+e^{-12(z-1/2)}}.$$

Independent censoring is generated as $C_i = \min(20, \widetilde{C}_i)$ with $\widetilde{C}_i \sim \text{Exp}(0.1)$. We use $n \in \{200, 500, 1000, 2000\}$ and 1000 Monte Carlo replications per setting. The same sample sizes, treatment process, treatment effect, and censoring mechanism are used in both baseline-risk scenarios.

5.2 Methods for Comparison

We compare TV-CSL with two single-stage Cox estimators, *S-Lasso* and *S-Cox*, where "single-stage" means that no propensity model is used. *S-Lasso* fits the outcome model with a cross-validated lasso penalty, whereas *S-Cox* fits the same model structure using an unpenalized Cox model.

All methods use two specifications for the baseline-risk function η_0: a linear specification and a complex specification containing natural splines, squared terms, and pairwise interactions. These are fitted model classes, distinct from the two DGPs described above. Thus, we consider two DGPs crossed with two outcome-model specifications. All methods use a linear model for τ_0. TV-CSL uses *S-Lasso* as the outcome nuisance model.

Furthermore, TV-CSL uses one of three propensity specifications:

- *Intercept-only* uses a constant propensity and is deliberately misspecified.
- *Marginal propensity* is the direct staggered-adoption extension of Gao and Hastie (2025) in Equation 15. Among subjects with $\Delta_i = 1$, it fits a Cox model for treatment initiation with response $\text{Surv}\{\min(A_i, U_i), \mathbf{1}(A_i < U_i)\}$ and predictors X_i.

- *Time-varying propensity* estimates our $a_0(t, x)$ directly by fitting the event regression $\operatorname{logit} a_0(t, x) = s(t) + x_1 + x_2 + x_3$ on $\{(U_i, X_i, W_i(U_i)) : \Delta_i = 1\}$.

Thus, TV-CSL has six fitted variants per DGP: two outcome specifications crossed with three propensity specifications.

We do not include causal survival forests (Cui et al., 2023) or other survival learners for heterogeneous effects. These methods require treatment to be fixed at baseline and target effects on restricted mean survival time or survival probabilities. Applying them here would require either classifying units by eventual adoption, which Section 6.2 shows to be biased, or using a landmark design with a different estimand. Such a comparison would therefore confound the choice of estimand with the handling of treatment timing.

We report mean squared error (MSE), $\|\hat{\beta} - \beta_0\|^2$, and empirical 95% coefficient-wise coverage, averaged over the three treatment-effect coordinates. For TV-CSL, coverage is calculated using both the information-based and sandwich standard errors from Section 4. The single-stage estimators use the usual Cox information standard errors.

5.3 Results

The main results are presented in Figures 3 and 4. In these figures, we omit S-Lasso because S-Cox performs strictly better. Appendix C.1 reports results for every fitted outcome specification and sample size for all methods.

We first discuss MSE. Figure 3 shows the full trend across all sample sizes. When η_0 is linear, direct Cox estimation performs better than the TV-CSL methods across all sample sizes. This is expected: when all models are correctly specified, the estimators have the same asymptotic target, while the simpler model can perform better in finite samples. When η_0 is non-linear, TV-CSL benefits from the DML orthogonality property

and performs better once the sample size is sufficiently large. At smaller sample sizes, Cox performs better because TV-CSL requires estimation of additional nuisance functions.

We next discuss coverage as shown in Figure 4. Under linear η_0, all procedures achieve coverage close to the nominal level except TV-CSL with the intercept-only propensity. Under non-linear η_0, only TV-CSL with the marginal or time-varying propensity achieves coverage close to the nominal level. The poor coverage of Cox is because the outcome nuisance converges too slowly, whereas the poor coverage of TV-CSL with the intercept-only propensity results from misspecification of the propensity model, i.e., the conditions of Proposition 4 are not satisfied. For both Cox and TV-CSL with the intercept-only propensity, coverage deteriorates as the sample size increases because the estimators converge more precisely to a biased target.

In Figure 4, coverage is based on the information-based variance estimator. We find that information-based TV-CSL intervals are closer to nominal coverage than the empirical sandwich intervals at smaller sample sizes, while the two become closer as n increases.

Among the TV-CSL specifications, marginal and time-varying centering give similar MSE and coverage, whereas the intercept-only specification is clearly worse because of propensity-score misspecification. In theory, time-varying centering is the centering implied by the orthogonality condition, whereas marginal centering is only an approximation. Their similar performance in the main simulation appears to arise because the non-linear η_0 setting is sufficiently difficult that the difference between the two propensity specifications has relatively little impact. In Appendix C.2, we therefore consider a simulation designed to isolate the centering difference. There, exact time-varying centering has smaller bias and better coverage than marginal centering, and the difference increases with the magnitude of the treatment effect.

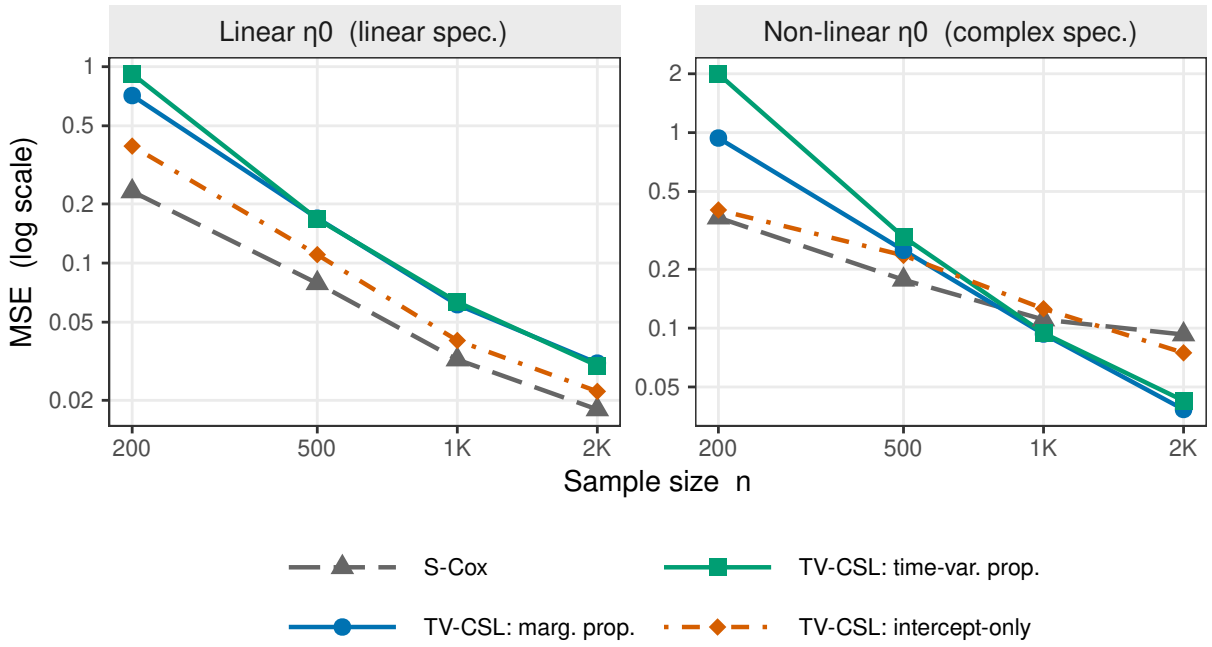


Figure 3: Mean squared error by method and sample size, using the linear fitted specification under the linear DGP and the complex fitted specification under the non-linear DGP. Under linear η_0, the direct single-stage fits are most accurate among the non-degenerate propensity estimators. Under non-linear η_0, TV-CSL with a non-degenerate estimated propensity has the smallest MSE at the larger sample sizes.

6 Data Analysis: Copilot Adoption

Generative-AI tools are increasingly available in professional services, but trying a tool once does not imply that it becomes part of regular work. We use lawyer-level data from a global law firm to estimate whether, and by how much, first observed use of the firm's internally developed AI assistant, changes the rate of becoming a regular Copilot user. We also examine whether this effect differs between junior and senior lawyers. This question is important because firms increasingly deploy several complementary AI tools, and evidence

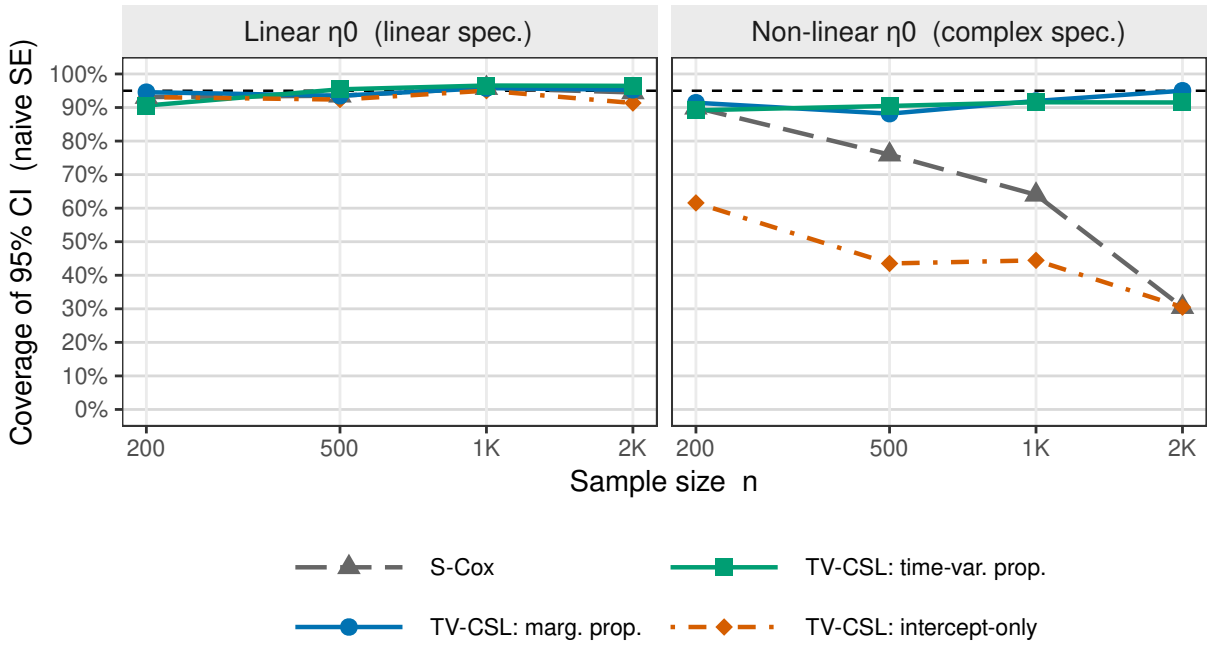


Figure 4: Empirical 95% coverage using information-based standard errors, averaged over the three treatment-effect coefficients. The dashed line marks 0.95. Under non-linear η_0, near-nominal coverage is obtained only by TV-CSL with the marginal or direct time-varying propensity. For both Cox and TV-CSL with the intercept-only propensity, coverage deteriorates as the sample size increases because the estimators converge more precisely to a biased target.

that one tool encourages broader adoption of another can inform investments in product integration, training, and employee adoption.

Time is measured in weeks from 6 October 2024, when Copilot telemetry begins. We define regular Copilot use as the first week with at least five recorded Copilot actions, or prompts, across applications such as Outlook and PowerPoint. The threshold was chosen through an exploratory examination of weekly usage patterns. It is an operational measure of engagement: reaching it once does not imply sustained use, and the available action

counts do not distinguish the quality or usefulness of those actions. Lawyers who do not reach the threshold during observable follow-up are censored.

The analysis has two goals. First, we show how treating eventual the AI assistant use as a baseline treatment, rather than accounting for its adoption date, changes the estimates. Second, using the AI assistant use as a time-varying treatment, we compare the time-varying Cox model with TV-CSL. Throughout this section, the main effect measure we report is the hazard ratio. A hazard ratio of h means that lawyers who had started using the AI assistant were approximately h times as likely, at any given time, to become regular Copilot users as comparable lawyers who had not yet started using the AI assistant.

6.1 Study Population, Variables for the Analysis and Summary Statistics

The unit of analysis is a lawyer. the AI assistant use is the treatment, and regular Copilot use is the event. Lawyers with no observed the AI assistant use after time zero remain untreated during follow-up. When initiation and regular Copilot use occur in the same week, the primary convention records the event as untreated. Follow-up ends at week 81, with earlier censoring at the last week the lawyer appears in the Copilot telemetry, including weeks with no Copilot use. We assume this censoring is non-informative given the adjustment covariates; telemetry presence approximates employment but can end earlier for lawyers who become inactive in the system.

The adjustment covariates are rank, office, practice group, weekly working hours, and prior the AI assistant use. Rank records a lawyer's professional level, office records the lawyer's location, and practice group records the lawyer's area of legal practice. Weekly working hours are averaged over each lawyer's first four observed weeks. We use seniority

as the treatment-effect modifier. Seniority is derived from rank: Partners, Counsel, Senior Associates, and Mid-Level Associates are classified as senior, whereas Junior Associates and Trainees are classified as junior.

Of the whole 3,874 lawyers in the dataset, we exclude 551 who were already regular Copilot users at the start of follow-up, because their transition to regular use is not observed. This leaves 3,323 lawyers. Table 1 reports the summary statistics for the 3,323 lawyers. We report only a subset of the summary statistics to prevent the company from being identified.

For the primary analysis, we further exclude 1,296 lawyers who used the AI assistant between 23 June 2024, when the AI assistant telemetry begins, and time zero, so that the analysis includes only new the AI assistant users. We also exclude one lawyer who became a regular user in their entry week and so contributes no time at risk. The primary population is 2,026 lawyers, of whom 1,891 became regular Copilot users during follow-up. As a sensitivity analysis, we analyse all 3,322 eligible lawyers (the 3,323 minus the same one lawyer). There, we use prior the AI assistant use as a binary adjustment variable in fitting nuisance models.

Among the 1,891 regular-use events in this subgroup, 729 (38.6%) occurred after the AI assistant use and 1,162 (61.4%) occurred before or in the same week as the AI assistant use. Both treatment states were represented within each event-time and seniority group. This means the treatment variable in the dataset is proper for time-varying effect estimation. When initiation and regular Copilot use occur in the same week, the primary convention records the event as untreated.

6.2 Effect of Ignoring Treatment Time

In this section, we compare two Cox models using the same cohort. The first treats eventual observed the AI assistant use as a fixed treatment variable. A subject is classified as treated if they receive treatment at any point, regardless of when treatment occurs. A control is a subject who receives no treatment before the event of becoming a regular user. The second uses the time-varying treatment process in Equation 4, with the AI assistant initiation as adoption.

Table 2 shows that the two models give very different answers. Both models find a significant effect for senior and junior lawyers, but the magnitudes are different: compared to the time-varying model, in the fixed treatment model, seniors have a 1.29 times effect in terms of hazard ratio (point estimate: 2.11 divided by 1.63), while juniors have a 1.69 times effect (point estimate: 2.14 divided by 1.27). Most importantly, the difference between junior and senior is significant in the time-varying model (ratio 0.78, $p = 0.015$) but not in the fixed treatment model (ratio 1.01, $p = 0.934$).

6.3 TV-CSL Results and Sensitivity Analyses

In this section, we apply three estimators to the data: the time-varying Cox model from the previous section and two TV-CSL specifications, one using the Gao–Hastie propensity in Equation 15 and the other using the time-varying propensity in Equation 13. Nuisance models are cross-fitted at the lawyer level.

The results are reported in Table 3. The point estimates are similar across the two TV-CSL specifications, as expected, although their confidence intervals differ. Qualitatively, both specifications suggest a positive association between the AI assistant use and becoming a regular Copilot user, with a larger estimated effect among senior lawyers. Quantitatively,

we focus on TV-CSL with the time-varying propensity because the simulation study indicates that this specification performs well. Each week, senior lawyers who had started using the AI assistant were about 1.5 times as likely to become regular Copilot users as comparable non-users (95% CI: 1.3–1.8), while junior lawyers were about 1.2 times as likely (95% CI: 1.0–1.5).

We also conducted sensitivity analyses, reported in Table 10 in the Appendix. The qualitative conclusion is unchanged when we vary how same-week adoption and event times are handled, although the estimated effects become larger. If lawyers who first used the AI assistant in the same week that they became regular Copilot users are counted as having used the AI assistant first, the estimated hazard ratios increase to approximately 1.9 for senior lawyers and 1.8 for junior lawyers. The qualitative conclusions are also similar across the two analysis populations considered.

If the measured adjustment variables are sufficient to control confounding and the identifying assumptions in Section 4 hold, the time-varying TV-CSL estimates have a causal interpretation: first observed the AI assistant use increases the instantaneous rate of becoming a regular Copilot user for both senior and junior lawyers. One possible explanation is that the AI assistant provides lawyers with an accessible introduction to AI-assisted work, making subsequent regular Copilot use easier.

7 Conclusion

We have connected a staggered-adoption causal design to a survival model for a once-observed event time. Under no anticipation, invariance to adoption history, unconfounded adoption time, and overlap, the time-dependent Cox contrast has a causal interpretation as a heterogeneous log hazard ratio. For estimation, the key object is not the ordinary adop-

tion propensity but the probability of prior treatment among subjects who fail at the current event time. Centering by this risk-set modified propensity yields a Neyman-orthogonal partial-likelihood score, a second-order nuisance remainder, and influence-function-based pointwise inference. More broadly, the hazard-weighted change of measure explains why treatment centering in survival analysis must account for both the contemporaneous risk set and the relative event hazards. This perspective may open the door to developing orthogonal estimating equations for other causal survival problems.

The simulations give a practical qualification to the theory. When baseline risk is simple, direct Cox estimation is more precise and there is little reason to estimate another nuisance. When baseline risk is complex, the cost of orthogonalization is eventually outweighed by its protection against first-stage error. In that setting, valid event-time centering is also important for inference: a deliberately misspecified constant propensity has both higher MSE and severe undercoverage.

The framework applies whenever treatment is adopted once and both treated and not-yet-treated subjects can experience the event at the same analysis times. The Copilot study shows why this design condition must be checked before estimation, while the Stanford Heart Transplant data in Appendix E provide the classical setting in which events may occur both before and after adoption.

7.1 Limitations and Future Work

The causal interpretation depends on stronger conditions than the statistical Cox fit alone. SUTVA may fail when an intervention has materially different versions, such as transplant procedures that vary across surgeons or hospitals. Unconfoundedness requires measurement of the variables that jointly determine adoption and the event. In transplantation, these include disease severity, comorbidities, organ compatibility, geography, and allocation rules

(Cascino et al., 2022; Khush et al., 2013; Parker et al., 2019; Power et al., 2024). Overlap can also be poor when such rules nearly determine who is treated at a given time. These are design limitations and cannot be repaired by orthogonalization.

No anticipation and invariance to adoption history are also substantive restrictions. The latter excludes duration-dependent effects. For example, mortality risk may increase immediately after transplantation because of surgical complications or acute rejection and then decline among patients who survive the early period (Lipkova et al., 2022). Such a pattern requires treatment-time or duration-specific effects rather than the single post-adoption hazard ratio used here. The present framework also does not address time-dependent confounders affected by prior treatment. More generally, if the treatment effect varies with calendar time or time since adoption, the proportional-hazards restriction is misspecified and the resulting HLHR should be interpreted as a working summary; extending TV-CSL to a time-varying contrast $\tau(t, x)$ is left for future work.

Our finite-dimensional, prespecified basis $p(x)$ is another limitation. It permits interpretable transformations and interactions but does not adapt non-parametrically to an unknown effect surface. Future work could develop an information-weighted projection target, uniform confidence bands, and tests of which subgroups benefit, extending related work for continuous outcomes (Dukes et al., 2024). It would also be useful to study sensitivity analysis for unmeasured adoption-time confounding and diagnostics for event-risk overlap.

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A Direct estimation of the event propensity

This section derives three equivalent interpretations of the event propensity $a_0(t, x)$: the treatment probability among individuals experiencing an event at time t, the treatment probability under the hazard-weighted risk-set measure P_{R_t}, and an adjustment of the ordinary adoption propensity for survival selection and relative event hazards. We then describe its direct estimation and derive the analytical oracle used in the simulations.

A.1 Event propensity and change of measure

We establish the observable characterization in Equation (13) and its equivalence to the hazard-weighted measure P_{R_t}.

For $h > 0$, write $\Delta_h N(t) = N(t+h) - N(t)$. Under the counting-process intensity, at regular event times,

$$\begin{aligned} &P\{W(t) = w, \Delta_h N(t) = 1 \mid X = x\} \\ &\quad = R_w(t,x)\lambda_0(t)e^{\eta_w(x)}h + o(h). \end{aligned} \tag{22}$$

Here $o(h)/h \to 0$ as $h \downarrow 0$; the expansion assumes local continuity and excludes simultaneous adoption and failure jumps.

Applying Bayes' rule and cancelling the common factor $\lambda_0(t)h$ gives, whenever $\lambda_0(t) > 0$,

$$\begin{aligned} a_0(t,x) &= \lim_{h \downarrow 0} P\{W(t) = 1 \mid \Delta_h N(t) = 1, X = x\} \\ &= \frac{R_1(t,x)e^{\eta_1(x)}}{\sum_{w=0}^{1} R_w(t,x)e^{\eta_w(x)}}. \end{aligned} \tag{23}$$

This proves Equation (13).

The equivalence extends to the joint distribution of pre-event characteristics. For any measurable covariate set B,

$$\begin{aligned} &\lim_{h \downarrow 0} P\{W(t) = w, X \in B \mid \Delta_h N(t) = 1\} \\ &\quad = \frac{E[Y(t)e^{\eta_{W(t)}(X)}\mathbf{1}\{W(t) = w, X \in B\}]}{c_t} \\ &\quad = P_{R_t}\{W(t) = w, X \in B\}. \end{aligned}$$

Thus the distribution of $(X, W(t))$ under P_{R_t} is the limiting distribution of these characteristics among subjects experiencing an event at time t. The equivalence concerns pre-event characteristics, not the entire observed record conditional on failure.

A.2 Relationship with the ordinary adoption propensity

Proof of event-time overlap. By conditioning on adoption status,

$$R_1(t,x) = \pi_t(x)P(U \geq t \mid A < t, X = x),$$
$$R_0(t,x) = \{1 - \pi_t(x)\}P(U \geq t \mid A \geq t, X = x).$$

Consequently,

$$\frac{a_0(t,x)}{1 - a_0(t,x)} = \frac{\pi_t(x)}{1 - \pi_t(x)}\rho_t(x)e^{\tau_0(x)},$$

which establishes the logit identity in the event-time overlap lemma.

Under Assumption 3,

$$\epsilon \leq \rho_t(x) \leq \epsilon^{-1}$$

on $[t_0, \bar{t}]$. Together with $\|\tau_0\|_\infty < \infty$, this gives

$$\epsilon' \leq a_0(t,x) \leq 1 - \epsilon',$$

where

$$\epsilon' = \{1 + \exp(L)\}^{-1}, \qquad L = \log\frac{1-\epsilon}{\epsilon^2} + \|\tau_0\|_\infty.$$

This completes the proof. □

A.3 Direct estimation from observed failures

Equation (13) suggests estimating $a_0(t,x)$ through the binary regression

$$E\{W(U) \mid U = t, X = x, \Delta = 1\} = a_0(t,x).$$

Within each training fold, retain one observation per observed failure:

$$\mathcal{D}_{\text{event}}^{(-k)} = \{(U_i, X_i, W_i(U_i)) : i \in I_k^c, \Delta_i = 1\}.$$

In our simulations, we fit the logistic generalized additive model

$$\operatorname{logit} a(t, x) = s(t) + x^\top \alpha, \tag{24}$$

where $s(t)$ is a spline in event time with basis dimension five.

The fitted model is evaluated for all members of each held-out risk set, not only those who experience an event. Although the additive specification shares information across event times, it may miss time–covariate interactions; the analytical oracle below illustrates this limitation.

A.4 Analytical oracle for the simulation DGP

For the simulation design, treatment adoption follows

$$A \mid X = x \sim \text{Exp}\{r(x)\}, \qquad r(x) = e^{x_2 + x_3},$$

and the event hazard satisfies

$$\lambda_0(t) = t, \qquad \Lambda_0(t) = t^2/2, \qquad \tau_0(x) = x_1 + x_2 + x_3.$$

We calculate the two treatment-specific risk-set probabilities analytically. Independent censoring contributes a common factor $G_C(t) = P(C \geq t)$, which cancels from $a_0(t, x)$.

Subjects not yet treated at t. An untreated subject at time t must have $A \geq t$ and survive under the untreated hazard until t. Thus

$$P(A \geq t \mid X = x) = e^{-r(x)t},$$
$$P(T \geq t \mid A \geq t, X = x) = \exp\left\{-\tfrac{1}{2}e^{\eta_0(x)}t^2\right\}.$$

Consequently,

$$R_0(t,x) = G_C(t)\exp\left\{-r(x)t - \tfrac{1}{2}e^{\eta_0(x)}t^2\right\}. \tag{25}$$

Subjects already treated at t. For an individual adopting at time $A = s < t$, the event hazard is untreated before s and treated between s and t. The conditional survival probability is

$$P(T \geq t \mid A = s, X = x)$$
$$= \exp\left\{-\tfrac{1}{2}e^{\eta_0(x)}s^2 - \tfrac{1}{2}e^{\eta_1(x)}(t^2 - s^2)\right\}.$$

Integrating over the adoption-time density $r(x)e^{-r(x)s}$ gives

$$R_1(t,x) = G_C(t)r(x)e^{-\frac{1}{2}e^{\eta_1(x)}t^2}$$
$$\times \int_0^t \exp\left[-r(x)s + \tfrac{1}{2}e^{\eta_0(x)}\{e^{\tau_0(x)} - 1\}s^2\right] \mathrm{d}s. \tag{26}$$

Define

$$d(x) = \tfrac{1}{2}e^{\eta_0(x)}\{e^{\tau_0(x)} - 1\}, \qquad I(t,x) = \int_0^t e^{-r(x)s + d(x)s^2}\,\mathrm{d}s.$$

Equations (25) and (26) imply

$$\frac{R_1(t,x)}{R_0(t,x)} = r(x)e^{r(x)t - d(x)t^2}I(t,x).$$

Finally,

$$\operatorname{logit} a_0(t,x) = \log \frac{R_1(t,x)}{R_0(t,x)} + \tau_0(x).$$

Substituting $r(x) = e^{x_2+x_3}$ and $\tau_0(x) = x_1 + x_2 + x_3$ yields

$$\boxed{\begin{aligned}\operatorname{logit} a_0(t,x) &= x_1 + 2x_2 + 2x_3 \\ &\quad + r(x)t - d(x)t^2 + \log I(t,x).\end{aligned}} \tag{27}$$

The formula applies to both simulation scenarios, with η_0 set to its linear or nonlinear specification. The integral $I(t,x)$ is evaluated numerically.

The oracle formula demonstrates why the additive GAM is generally only an approximation: both $d(x)t^2$ and $\log I(t,x)$ contain time–covariate interactions absent from the additive working model.

B Proofs for the DML results

B.1 Technical regularity conditions

For a measurable scalar or vector function $f(t, X, W(t))$, define the event-weighted norm

$$\|f\|_{2,\mu}^2 = \int_0^{\bar{t}} c_t E_{R_t}[\|f(t,X,W(t))\|^2]\, d\Lambda_0(t). \tag{28}$$

This integrates the squared error under the event-risk distribution across time.

Condition 3 (Technical DML regularity)**.** (i) β_0 is an interior point of a compact convex parameter space $\mathcal{B}$, and $\Psi(\beta, a_0, \nu_0)$ has a unique zero on $\mathcal{B}$.

(ii) The nuisance-weighted risk-set denominators remain positive in a neighborhood of

the truth. The expectations defining the population score admit integrable envelopes permitting differentiation under expectations and time integrals.

The second derivatives $D^2_{aa}\Psi$ and $D^2_{a\nu}\Psi$ are uniformly bounded as bilinear operators under the norm in Equation (28), along admissible nuisance paths.

(iii) The number of folds K is fixed, with n_k/n bounded away from zero. Furthermore,

$$\sup_x \|p(x)\| < \infty, \qquad \sup_{x,w} |\eta_w(x)| < \infty.$$

With probability approaching one, the cross-fitted propensity estimates belong to $[0, 1]$, and the cross-fitted $\hat{\nu}$ are uniformly bounded.

B.2 Proof of Proposition 2

Recall the measure P_{R_t} in Equation (11) and the centered regressor $q_0(t)$ defined in Section 4.2. By the definition of a_0,

$$E_{R_t}[q_0(t) \mid X] = 0, \qquad E_{R_t}[q_0(t)] = 0. \tag{29}$$

The measure P_{R_t} is held fixed at the true event intensity throughout the nuisance perturbations. The measure is fixed at the true hazard throughout the following perturbations.

Under the assumed counting-process intensity, for any integrable predictable function f,

$$E \int_0^{\bar{t}} f(t)\, dN(t) = \int_0^{\bar{t}} c_t E_{R_t}[f(t)]\, d\Lambda_0(t).$$

Thus the population score can be written as

$$\Psi(\beta, a, \nu) = \int_0^{\bar{t}} c_t \left\{ E_{R_t}[q_a(t)] - \bar{q}(t; \beta, a, \nu) \right\} d\Lambda_0(t). \tag{30}$$

We first express the risk-set mean under P_{R_t}. Recall that

$$\nu_0(t, X) + q_0(t)^\top \beta_0 = \eta_{W(t)}(X).$$

For arbitrary a and ν, define

$$H_{a,\nu}(t, X) = \exp\left\{ \nu(t, X) - \nu_0(t, X) - [a(t, X) - a_0(t, X)]\tau_0(X) \right\}.$$

Since

$$e^{\nu(t,X) + q_a(t)^\top \beta_0} = e^{\eta_{W(t)}(X)} H_{a,\nu}(t, X),$$

the population risk-set mean satisfies

$$\bar{q}(t; \beta_0, a, \nu) = \frac{E_{R_t}[H_{a,\nu}(t, X) q_a(t)]}{E_{R_t}[H_{a,\nu}(t, X)]}. \tag{31}$$

Invariance to ν. Fix $a = a_0$ and let ν be any admissible nuisance function. Then $q_a = q_0$ and

$$H_{a_0,\nu}(t, X) = e^{\nu(t,X) - \nu_0(t,X)}.$$

This factor depends only on (t, X). By Equation (29),

$$\begin{aligned} &E_{R_t}[H_{a_0,\nu}(t, X) q_0(t)] \\ &\quad = E_{R_t}\left[H_{a_0,\nu}(t, X) E_{R_t}[q_0(t) \mid X] \right] = 0. \end{aligned}$$

Equation (31) therefore gives

$$\bar{q}(t; \beta_0, a_0, \nu) = 0.$$

Since $E_{R_t}[q_0(t)] = 0$, substituting into Equation (30) yields

$$\Psi(\beta_0, a_0, \nu) = 0$$

for every admissible ν.

In particular, differentiation at ν_0 gives

$$D_\nu \Psi(\beta_0, a_0, \nu_0)[h_\nu] = 0.$$

Orthogonality with respect to a. Consider the joint perturbation

$$a_s = a_0 + sh_a, \qquad \nu_s = \nu_0 + sh_\nu,$$

where h_a and h_ν are arbitrary admissible directions and s is sufficiently close to zero.

Write

$$q_s(t) = q_{a_s}(t) = q_0(t) - sh_a(t, X)p(X)$$

and

$$H_s(t, X) = H_{a_s, \nu_s}(t, X) = \exp\left\{s[h_\nu(t, X) - h_a(t, X)\tau_0(X)]\right\}.$$

First, the derivative of the observed-event component in Equation (30) is

$$\left.\frac{d}{ds} E_{R_t}[q_s(t)]\right|_{s=0} = -E_{R_t}[h_a(t, X)p(X)]. \tag{32}$$

Next, by Equation (31),

$$\bar{q}(t;\beta_0,a_s,\nu_s)=\frac{E_{R_t}[H_s(t,X)q_s(t)]}{E_{R_t}[H_s(t,X)]}.$$

At $s=0$, we have $H_0=1$ and

$$E_{R_t}[H_0q_0(t)]=E_{R_t}[q_0(t)]=0.$$

Differentiating the numerator gives

$$\begin{aligned}&\frac{d}{ds}E_{R_t}[H_s(t,X)q_s(t)]\Big|_{s=0}\\&\quad=-E_{R_t}[h_a(t,X)p(X)]\\&\qquad+E_{R_t}\left[q_0(t)\{h_\nu(t,X)-h_a(t,X)\tau_0(X)\}\right].\end{aligned}$$

The last expectation is zero because the factor multiplying $q_0(t)$ depends only on (t,X) and $E_{R_t}[q_0(t)\mid X]=0$. Therefore,

$$\frac{d}{ds}E_{R_t}[H_sq_s]\Big|_{s=0}=-E_{R_t}[h_a(t,X)p(X)].$$

Since the numerator of the quotient is zero at $s=0$ and its denominator equals one, the quotient rule gives

$$\frac{d}{ds}\bar{q}(t;\beta_0,a_s,\nu_s)\Big|_{s=0}=-E_{R_t}[h_a(t,X)p(X)]. \tag{33}$$

Equations (32) and (33) are identical. Their contributions therefore cancel in the pop-

ulation score:

$$\frac{d}{ds}\Psi(\beta_0, a_s, \nu_s)\Big|_{s=0} = 0.$$

Taking $h_\nu = 0$ gives

$$D_a\Psi(\beta_0, a_0, \nu_0)[h_a] = 0.$$

Together with the result for ν, this establishes Neyman orthogonality with respect to both nuisance functions. □

B.3 Proof of Proposition 3

Write $g = (a, \nu)$ and $g_0 = (a_0, \nu_0)$. For each fold k, let $\hat{g}_k = (\hat{a}^{(-k)}, \hat{\nu}^{(-k)})$ and $\kappa_k = n_k/n$.

Step 1: Population nuisance remainder. Let $\delta a = a - a_0$, $\delta\nu = \nu - \nu_0$, and $H = H_{a,\nu}$ as defined in the proof of Proposition 2.

Using $q_a = q_0 - \delta a\, p$ and $E_{R_t}[q_0(t) \mid X] = 0$, Equation (30) gives the exact identity

$$\Psi(\beta_0, a, \nu) = \int_0^{\bar{t}} c_t \frac{\mathrm{Cov}_{R_t}\{\delta a\, p, H\}}{E_{R_t}[H]}\, d\Lambda_0(t), \tag{34}$$

where covariances are evaluated at time t and interpreted componentwise.

At the truth, $H = 1$ and

$$DH[h_a, h_\nu] = h_\nu - h_a\tau_0.$$

Differentiating Equation (34) therefore yields

$$
\begin{aligned}
&D^2_{aa}\Psi[h_a, k_a] \\
&\quad = -\int_0^{\bar{t}} c_t \left[\mathrm{Cov}_{R_t}(h_a p, k_a \tau_0) + \mathrm{Cov}_{R_t}(k_a p, h_a \tau_0)\right] d\Lambda_0(t), \qquad (35)\\
&D^2_{a\nu}\Psi[h_a, h_\nu] = \int_0^{\bar{t}} c_t \,\mathrm{Cov}_{R_t}(h_a p, h_\nu)\, d\Lambda_0(t), \qquad (36)\\
&D^2_{\nu\nu}\Psi[h_\nu, k_\nu] = 0. \qquad (37)
\end{aligned}
$$

All derivatives are evaluated at (β_0, g_0).

By Proposition 2, $\Psi(\beta_0, a_0, \nu) = 0$ for every admissible ν. Applying the fundamental theorem of calculus first along the a direction and then to its first derivative gives

$$
\begin{aligned}
&\Psi(\beta_0, a, \nu) - \Psi(\beta_0, g_0) \\
&= \int_0^1 \int_0^s D^2_{aa}\Psi(\beta_0, a_0 + u\delta a, \nu)[\delta a, \delta a]\, du\, ds \\
&\quad + \int_0^1 D^2_{a\nu}\Psi(\beta_0, a_0, \nu_0 + v\delta\nu)[\delta a, \delta\nu]\, dv.
\end{aligned}
$$

The bilinear derivative bounds in Condition 3(ii) and Cauchy–Schwarz imply

$$
\|\Psi(\beta_0, a, \nu) - \Psi(\beta_0, g_0)\| \le C\left(\|\delta a\|^2_{2,\mu} + \|\delta a\|_{2,\mu}\|\delta\nu\|_{2,\mu}\right).
$$

Applying this bound fold by fold gives

$$
B_n := \sum_{k=1}^{K} \kappa_k \Psi(\beta_0, \hat{g}_k) = O_p\{r^2_{a,n} + r_{a,n} r_{\nu,n}\}. \qquad (38)
$$

Step 2: Oracle empirical score. For any function f, write

$$P_{n,k} f = \frac{1}{n_k} \sum_{i \in I_k} f(O_i).$$

Define the oracle empirical risk-set expectation

$$\langle f \rangle_{n,k,t} = \frac{P_{n,k}[Z_t f(t)]}{P_{n,k} Z_t}, \qquad Z_t = Y(t) e^{\eta_{W(t)}(X)}.$$

Let

$$\bar{q}^0_{n,k}(t) = \langle q_0(t) \rangle_{n,k,t}.$$

The oracle fold score is

$$\Psi^0_{n,k} = P_{n,k} \int_0^{\bar{t}} \{q_0(t) - \bar{q}^0_{n,k}(t)\} \, dN(t).$$

Using the Doob–Meyer decomposition

$$dN_i(t) = Z_{i,t} \, d\Lambda_0(t) + dM_i(t),$$

we obtain

$$\begin{aligned} \Psi^0_{n,k} = P_{n,k} \int_0^{\bar{t}} \{q_0(t) - \bar{q}^0_{n,k}(t)\} \, dM(t) \\ + \int_0^{\bar{t}} P_{n,k} \left[Z_t \{q_0(t) - \bar{q}^0_{n,k}(t)\} \right] d\Lambda_0(t). \end{aligned}$$

The second term is exactly zero because

$$P_{n,k}[Z_t q_0(t)] = P_{n,k} Z_t \, \bar{q}^0_{n,k}(t).$$

Thus

$$\Psi^0_{n,k} = P_{n,k} \int_0^{\bar{t}} q_0(t)\, dM(t) - \int_0^{\bar{t}} \bar{q}^0_{n,k}(t)\, P_{n,k} dM(t). \tag{39}$$

The population conditional centering identity gives $E[Z_t q_0(t)] = 0$. Independence across subjects and the bounded-envelope conditions imply

$$E \int_0^{\bar{t}} \|P_{n,k}[Z_t q_0(t)]\|^2 \, d\Lambda_0(t) = O(n_k^{-1}).$$

Since $P_{n,k} Z_t$ is uniformly bounded away from zero with probability approaching one,

$$\int_0^{\bar{t}} \|\bar{q}^0_{n,k}(t)\|^2 \, d\Lambda_0(t) = O_p(n_k^{-1}).$$

The empirical risk-set mean is predictable. Applying the martingale isometry, with localization on the event that risk-set denominators are bounded away from zero, gives

$$\int_0^{\bar{t}} \bar{q}^0_{n,k}(t)\, P_{n,k} dM(t) = O_p(n_k^{-1}).$$

Therefore,

$$\Psi^0_{n,k} = \frac{1}{n_k} \sum_{i \in I_k} \psi_i + O_p(n_k^{-1}),$$

where we recall in main text

$$\psi_i = \int_0^{\bar{t}} q_i^0(t)\, dM_i(t).$$

Aggregating over the fixed number of folds yields

$$\Psi_n(\beta_0, g_0) = \frac{1}{n} \sum_{i=1}^{n} \psi_i + O_p(n^{-1}) = O_p(n^{-1/2}). \tag{40}$$

Step 3: Cross-fitting remainder. We now account for nuisance estimation and the empirical risk-set ratios.

For each fold, let

$$\delta a_k = \hat{a}^{(-k)} - a_0, \qquad \delta \nu_k = \hat{\nu}^{(-k)} - \nu_0,$$

and define

$$H_k(t, X) = e^{\delta \nu_k(t,X) - \delta a_k(t,X)\tau_0(X)}.$$

Put

$$\rho_{k,n} = \|\delta a_k\|_{2,\mu} + \|\delta \nu_k\|_{2,\mu}.$$

Conditional on the training observations I_k^c, these functions are fixed and the held-out observations remain independent.

Write $\widehat{\Psi}_{n,k}(\beta, g)$ for the fold-specific empirical Cox score computed with nuisance functions g. Define

$$\mathcal{E}_n = \sum_{k=1}^{K} \kappa_k \Big[\widehat{\Psi}_{n,k}(\beta_0, \hat{g}_k) - \Psi(\beta_0, \hat{g}_k) \\ - \widehat{\Psi}_{n,k}(\beta_0, g_0) + \Psi(\beta_0, g_0) \Big]. \tag{41}$$

We establish the order of this term. Under the true intensity, the empirical score can be decomposed into a martingale component and a compensator component. For an arbitrary $g = (a, \nu)$, the latter is

$$D_{n,k}(g) = \int_0^{\bar{t}} P_{n,k} Z_t \left\{ \langle q_a(t) \rangle_{n,k,t} - \bar{q}_{n,k}(t; \beta_0, g) \right\} d\Lambda_0(t).$$

At $g = \hat{g}_k$, the same change-of-measure calculation as in Step 1 gives

$$D_{n,k}(\hat{g}_k) = \int_0^{\bar{t}} \frac{P_{n,k} Z_t}{\langle H_k \rangle_{n,k,t}} \Big[\mathrm{Cov}_{n,k,t}\{\delta a_k p, H_k\} - \mathrm{Cov}_{n,k,t}\{q_0, H_k\} \Big] d\Lambda_0(t), \tag{42}$$

where $\mathrm{Cov}_{n,k,t}$ denotes covariance under the empirical weighted risk-set distribution $\langle \cdot \rangle_{n,k,t}$.

At $g = g_0$, $H_k = 1$, and therefore

$$D_{n,k}(g_0) = 0$$

exactly.

By contrast, the population drift is

$$\Psi(\beta_0, \hat{g}_k) = \int_0^{\bar{t}} c_t \frac{\mathrm{Cov}_{R_t}\{\delta a_k p, H_k\}}{E_{R_t}[H_k]} \, d\Lambda_0(t),$$

because $\mathrm{Cov}_{R_t}(q_0, H_k) = 0$.

To compare these expressions, note that bounded predictors and the mean-value theorem give

$$|H_k(t, X) - 1| \leq C\{|\delta\nu_k(t, X)| + |\delta a_k(t, X)|\}.$$

Consequently,

$$\|H_k - 1\|_{2,\mu} \leq C\rho_{k,n}.$$

In particular, conditional on the training sample, the centered empirical averages of $Z_t q_0(t)\{H_k(t, X) - 1\}$ have integrated second moments bounded by $C n_k^{-1} \rho_{k,n}^2$. Their population expectations vanish by $E_{R_t}[q_0(t) \mid X] = 0$.

The remaining terms in Equation (42) are products and ratios of empirical weighted

averages. Their denominators are bounded away from zero, and their numerators satisfy the same integrated second-moment bounds. Expanding the ratios around their population counterparts and applying Cauchy–Schwarz gives

$$D_{n,k}(\hat{g}_k) - \Psi(\beta_0, \hat{g}_k) = O_p\left(n_k^{-1/2}\rho_{k,n} + n_k^{-1}\right). \tag{43}$$

For the martingale component, the difference between the fitted and oracle score integrands is

$$-\delta a_k(t, X)p(X) - \left\{\bar{q}_{n,k}(t; \beta_0, \hat{g}_k) - \bar{q}^0_{n,k}(t)\right\}.$$

The bounded-envelope and denominator conditions, together with the $L_2(\mu)$ nuisance rates, imply that its integrated squared intensity norm is $O_p(\rho^2_{k,n})$. The martingale isometry therefore bounds the difference between the fitted and oracle martingale components by

$$O_p(n_k^{-1/2}\rho_{k,n}).$$

Combining this bound with Equation (43) and summing over the fixed number of folds yields

$$\mathcal{E}_n = O_p\left\{n^{-1/2}(r_{a,n} + r_{\nu,n}) + n^{-1}\right\} = o_p(n^{-1/2}). \tag{44}$$

This establishes the stochastic remainder bound using the actual fold-restricted Cox risk sets, without imposing a Donsker condition on the nuisance learners.

Step 4: Consistency and rate. The bounded-envelope conditions and fold-conditional laws of large numbers give uniform convergence of the cross-fitted score:

$$\sup_{\beta \in \mathcal{B}} \left\| \widehat{\Psi}_n(\beta, \hat{g}) - \Psi(\beta, g_0) \right\| = o_p(1).$$

Here $\widehat{\Psi}_n(\beta, \hat{g})$ denotes the aggregate fold-specific score, rather than a score evaluated at one common nuisance estimate.

By compactness of $\mathcal{B}$, continuity of the population score, and uniqueness of its zero at β_0, any solution $\hat{\beta} \in \mathcal{B}$ to the cross-fitted estimating equation satisfies

$$\hat{\beta} \xrightarrow{p} \beta_0.$$

Differentiating the population score with respect to β at the truth gives

$$\begin{aligned} -\partial_\beta \Psi(\beta_0, g_0) &= \int_0^{\bar{t}} c_t \operatorname{Var}_{R_t}\{q_0(t)\}\, d\Lambda_0(t) \\ &= \int_0^{\bar{t}} c_t E_{R_t}[q_0(t) q_0(t)^\top]\, d\Lambda_0(t) = J, \end{aligned}$$

where the second equality uses $E_{R_t}[q_0(t)] = 0$.

The same fold-conditional laws of large numbers, applied to the risk-set second moments, give local uniform convergence of the empirical score derivative to its population counterpart. Continuity of the latter, together with $\hat{\beta} \to_p \beta_0$, therefore implies

$$-\int_0^1 \partial_\beta \widehat{\Psi}_n\{\beta_0 + s(\hat{\beta} - \beta_0), \hat{g}\}\, ds = J + o_p(1).$$

Applying the fundamental theorem of calculus to the estimating equation yields

$$\begin{aligned} 0 &= \widehat{\Psi}_n(\beta_0, \hat{g}) - \{J + o_p(1)\}(\hat{\beta} - \beta_0) \\ &= \Psi_n(\beta_0, g_0) + B_n + \mathcal{E}_n \\ &\quad - \{J + o_p(1)\}(\hat{\beta} - \beta_0). \end{aligned}$$

Since J is nonsingular, Equations (38), (40), and (44) imply

$$\|\hat{\beta} - \beta_0\| = O_p\left\{n^{-1/2} + r_{a,n}^2 + r_{a,n} r_{\nu,n}\right\}.$$

This proves Proposition 3. □

B.4 Proof of Proposition 4

Under the stated rate condition,

$$\sqrt{n}\, r_{a,n}(r_{a,n} + r_{\nu,n}) \longrightarrow 0,$$

Equation (38) implies

$$B_n = o_p(n^{-1/2}).$$

Moreover, Equation (44) gives

$$\mathcal{E}_n = o_p(n^{-1/2}).$$

The score expansion established in the proof of Proposition 3 therefore reduces to

$$0 = \Psi_n(\beta_0, g_0) - \{J + o_p(1)\}(\hat{\beta} - \beta_0) + o_p(n^{-1/2}).$$

Since J is nonsingular,

$$\sqrt{n}(\hat{\beta} - \beta_0) = J^{-1}\sqrt{n}\, \Psi_n(\beta_0, g_0) + o_p(1).$$

By Equation (40),

$$\sqrt{n}(\hat{\beta} - \beta_0) = J^{-1}\frac{1}{\sqrt{n}}\sum_{i=1}^{n} \psi_i + o_p(1), \tag{45}$$

where

$$\psi_i = \int_0^{\bar{t}} q_i^0(t)\, dM_i(t).$$

Under the correctly specified counting-process intensity, each M_i is a square-integrable martingale. Since q_i^0 is predictable,

$$E[\psi_i] = 0.$$

The martingale isometry gives

$$\begin{aligned} E[\psi_i \psi_i^\top] &= E \int_0^{\bar{t}} q_i^0(t) q_i^0(t)^\top \, d\langle M_i \rangle(t) \\ &= E \int_0^{\bar{t}} Y_i(t) e^{\eta W_i(t)(X_i)} q_i^0(t) q_i^0(t)^\top \, d\Lambda_0(t) \\ &= J. \end{aligned}$$

In particular, $\Sigma = E[\psi_i \psi_i^\top]$ is finite and positive definite under Condition 2(i). The vectors ψ_i are independent and identically distributed across subjects. Thus the multivariate central limit theorem yields

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi_i \rightsquigarrow N(0, \Sigma).$$

Combining this convergence with Equation (45) and applying Slutsky's theorem gives

$$\sqrt{n}(\hat{\beta} - \beta_0) \rightsquigarrow N(0, J^{-1} \Sigma J^{-1}).$$

This proves Proposition 4. □

Variance estimation. We briefly verify consistency of the plug-in covariance estimator in Section 4.

At the truth, the population risk-set mean of $q_0(t)$ vanishes. Hence, by Equation (20), the fold-conditional laws of large numbers and nuisance consistency give

$$\hat{J} \xrightarrow{p} \int_0^{\bar{t}} c_t E_{R_t}[q_0(t) q_0(t)^\top] \, d\Lambda_0(t) = J.$$

For each fold, the Breslow estimator satisfies the exact decomposition

$$d\hat{\Lambda}_{0,k}(t) = \frac{P_{n,k} dM(t)}{\hat{S}_k^{(0)}(t; \hat{\beta})} + \frac{P_{n,k} Z_t}{\hat{S}_k^{(0)}(t; \hat{\beta})} \, d\Lambda_0(t).$$

Under the bounded-envelope and risk-set stability conditions, nuisance consistency and $\hat{\beta} \to_p \beta_0$ imply

$$\frac{P_{n,k} Z_t}{\hat{S}_k^{(0)}(t; \hat{\beta})} \longrightarrow 1$$

in the integrated risk-set norm. The martingale term in the Breslow decomposition is $O_p(n_k^{-1/2})$ when integrated against uniformly bounded predictable score integrands.

Using this decomposition and the $L_2(\mu)$ convergence of $\hat{q}_i$ to q_i^0, the estimated subject-level influence scores satisfy

$$\frac{1}{n} \sum_{i=1}^{n} \|\hat{\psi}_i - \psi_i\|^2 = o_p(1).$$

Here replacing β_0 by $\hat{\beta}$ is justified by finite-dimensional local stochastic equicontinuity.

Consequently, Cauchy–Schwarz and the law of large numbers yield

$$\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^{n} \hat{\psi}_i \hat{\psi}_i^\top \xrightarrow{p} E[\psi_i \psi_i^\top] = \Sigma.$$

Since J is nonsingular, the continuous mapping theorem gives

$$\hat{\Omega} = \hat{J}^{-1}\hat{\Sigma}\hat{J}^{-1} \xrightarrow{p} J^{-1}\Sigma J^{-1} = \Omega.$$

C Additional Simulation Results

C.1 Complete simulation tables

See Tables 4 to 7.

C.2 Isolating the Effect of the Propensity Centering

In the main simulation, marginal and time-varying centering perform similarly under η_0^{NL}. This does not mean that the centering is unimportant. By Proposition 3, the bias of TV-CSL is of the order of the product of the propensity error and the outcome-model error. If the outcome learner is nearly exact, any reasonable centering gives small bias. If the outcome learner is far from η_0, only the exact $a_0(t, x)$ removes the resulting bias, and small differences between estimated centerings do not matter. Under η_0^{NL} the complex specification is in the second case: its fitted $\hat{\eta}_0$ has root-mean-squared error equal to 0.64 of the standard deviation of η_0^{NL}. At $n = 4000$ and $\beta_0 = (0, 2, 2, 2)^\top$, replacing $\hat{\eta}_0$ and $\hat{\eta}_1$ with the true functions reduces the slope bias of every centering to at most 0.014 in absolute value, with coverage between 0.93 and 0.95. With the estimated functions, the slope bias is -0.102 for marginal centering, -0.129 for time-varying centering, and -0.007 for the oracle.

To compare the centerings, we therefore need a baseline risk for which the outcome-

model error is neither negligible nor dominant. We use the family

$$\eta_0^{(w)}(x) = w\,\eta_0^{NL}(x) + (1-w)\,\eta_0^{S}(x), \qquad \eta_0^{S}(x) = \tfrac{3}{4}\{\sin x_1 + x_2^2/2 + x_1 x_3/2\}, \qquad w \in [0,1].$$

The function η_0^S is non-linear and non-additive, but the complex specification recovers it well. The weight w moves the baseline risk from η_0^S ($w = 0$) to η_0^{NL} ($w = 1$).

We choose w using only the first-stage outcome learner, before fitting any treatment-effect estimator. For each $w \in \{0, 0.25, 0.5, 0.75, 1\}$, we fit the complex specification at $n = 4000$ and compute the root-mean-squared error of $\hat{\eta}_0$ relative to the standard deviation of $\eta_0^{(w)}$, after centering both at their means. We fixed the target interval $[0.2, 0.4]$ in advance. Table 8 reports the relative errors. Only $w = 0.25$ falls in the interval.

The design is otherwise that of Section 5.1, with $\eta_0 = \eta_0^{(0.25)}$, $n = 4000$, and $\beta_0 = (0, m, m, m)^\top$ for $m \in \{0, 0.5, 1, 1.5, 2\}$. Each value of m uses the same 50 simulated datasets up to the event times, so differences across m are not driven by different covariate draws. We fit TV-CSL with the complex outcome specification and the marginal, time-varying, and oracle propensities of Section 5.2. The case $m = 0$ is a null check. When $\tau_0 = 0$, treatment does not change the hazard, so $a_0(t, x) = P(A < t \mid X = x)$ and the marginal propensity is correctly specified.

Table 9 reports the results. At $m = 0$ the three centerings are indistinguishable, as expected. As m grows, the bias of marginal centering grows, from -0.004 at $m = 0$ to -0.048 at $m = 2$, and its information-based coverage falls to 0.83. Time-varying centering reduces this bias by about one third at every $m \geq 1$. Paired over datasets, the reduction is 0.006, 0.011, and 0.017 at $m = 1$, 1.5, and 2, with z-statistics 2.7, 4.2, and 4.9. At $m = 2$, time-varying centering keeps coverage at 0.90. The oracle is unbiased at every m and shows how much of the remaining bias is due to error in the estimated a_0.

Time-varying centering does not reduce MSE. Its estimates are more variable than those of marginal centering, with empirical standard deviation 0.078 against 0.071 at $m = 2$, and this offsets the smaller bias. The gain from the correct centering is therefore in bias and coverage, which is what Proposition 3 concerns. No centering reaches 0.95, including the oracle at $m = 0$, so the remaining undercoverage comes from variance estimation rather than from the centering.

For comparison, the same experiment with $w = 1$, that is, under η_0^{NL}, gives similar bias for the two estimated centerings at every m; at $m = 2$ it is -0.107 for marginal and -0.120 for time-varying centering, while the oracle remains unbiased. This is the case in which the outcome-model error dominates.

D Data Analysis: Sensitivity Table

See Table 10.

E Additional Application: Stanford Heart Transplant Data

The Stanford Heart Transplant data come from the Stanford transplantation programme, which began in the 1960s. Patients entered the study with severe heart disease and waited for a compatible donor heart. The data follow survival from programme acceptance through the waiting and post-transplant periods and include age, prior surgery, and year of acceptance (Crowley and Hu, 1977). We use this small dataset as an additional illustration of the importance of updating treatment status at transplant time. In light of the simulation results and the sample size of 103, the direct time-dependent Cox analysis is the primary

analysis; the flexible comparison below is treated as a finite-sample diagnostic.

E.1 Summary statistics

The regressors are age at acceptance, an indicator of prior surgery, and year of acceptance measured from the start of the programme in 1967. We use time since programme acceptance as the survival time and include year of acceptance as a baseline covariate. This keeps time zero clinically comparable across patients while allowing outcomes to change over calendar time. Table 11 gives the summary statistics. The average age is 45.17 years, 16% of patients had prior surgery, and 69 of the 103 patients received a transplant. Figure 5 shows the variation in transplant time.

E.2 Effect of Ignoring Transplant Time

In this section, we compare two Cox models using all 103 patients. The first treats eventual transplant receipt as a fixed treatment variable: a patient is classified as treated if they receive a transplant at any point during follow-up, regardless of when it occurs. The second uses the time-varying treatment process in Equation 4, with transplant as adoption, so that each patient contributes untreated time while waiting and treated time after transplant. This can be seen as a replication of Crowley and Hu (1977).

When transplant and death or censoring occur on the same day, the transplant is recorded half a day earlier, so the event counts as treated; this affects one patient. Both models adjust for age, year of acceptance, and prior surgery. We use age and year of acceptance, both standardized, as treatment-effect modifiers. Prior surgery is an adjustment covariate only, because just 16 patients had prior surgery. As in Section 6, we report hazard ratios. Here, a hazard ratio of h means that patients who had received a transplant were

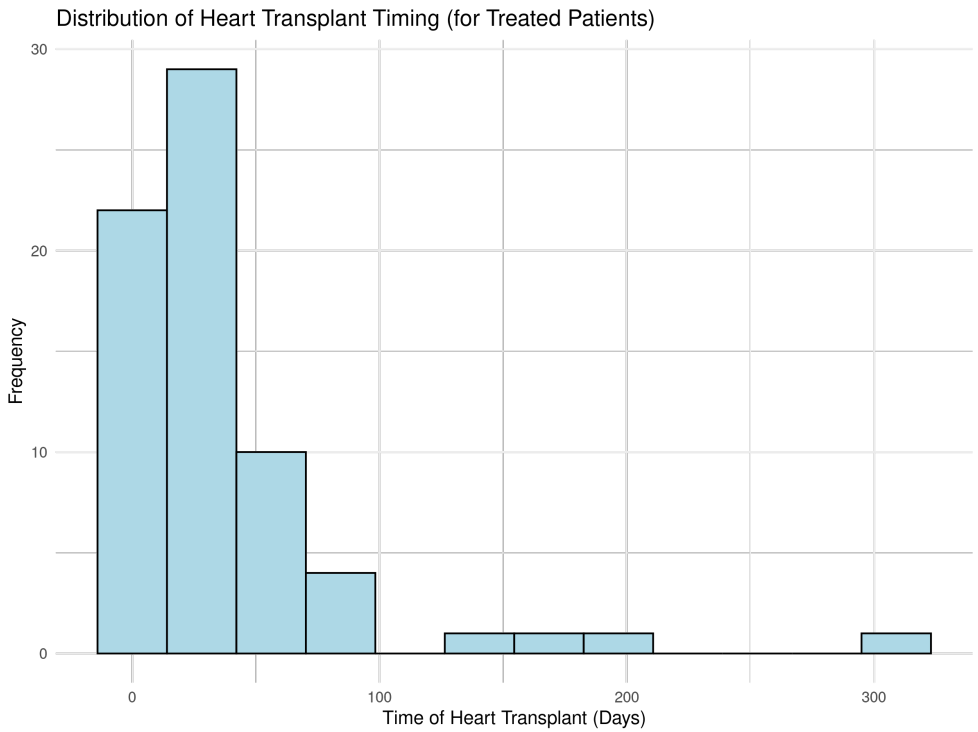


Figure 5: Distribution of Time to Heart Transplant

approximately h times as likely, at any given time, to die as comparable patients who were still waiting.

Table 12 shows that the two models give very different answers. When transplant time is ignored, transplant appears strongly protective: for a patient of average age accepted in an average year, the hazard ratio is 0.19 (log hazard ratio -1.648, $p < 0.001$). When transplant status is updated at the transplant date, the hazard ratio is 1.02 (log hazard ratio 0.020, $p = 0.951$), and there is no evidence that transplant changed mortality for this patient. The interactions also change: the age interaction changes sign, from -0.228 to 0.296, and the year interaction increases from 0.143 to 0.394, although neither is significant in either model. The fixed model counts the waiting period of eventual recipients as treated

time. Because a patient must survive the wait to receive a transplant, this period is event-free by construction, and the fixed model credits it to transplant. The difference between the two models is therefore consistent with immortal-time bias. The semi-synthetic study in the next section confirms this.

E.3 Semi-Synthetic Comparison at the Stanford Sample Size

In this section, we use a semi-synthetic study to answer two questions that the real data cannot answer on their own. First, is the difference in Table 12 due to immortal-time bias? Second, with only 103 patients, does TV-CSL improve on the time-varying Cox model? The true effects are unknown in the real data, so we construct data sets in which they are known but which otherwise resemble the Stanford data.

We keep each patient's real covariates and real censoring date, and redraw transplant and death times. Death times follow the time-varying Cox model fitted to the real data in the previous section, so the true effects are the time-varying estimates in Table 12: 0.019 for transplant, 0.298 for the age interaction, and 0.393 for the year interaction. Transplant times follow a Cox model for time to transplant, fitted to the real data with age, year of acceptance, and prior surgery. Both models use the baseline hazards estimated from the real data. The simulated data sets closely resemble the real data: on average, they contain 75.1 deaths and 68.4 transplants, compared with 75 and 69 in the real data. We generate 500 data sets and apply four estimators: the two Cox models from the previous section and the two TV-CSL specifications used in Section 6, with nuisance models cross-fitted over five folds at the patient level. For each estimator, we report the bias and the coverage of nominal 95% confidence intervals.

Table 13 reports the results. Ignoring transplant time is badly biased. On average, it underestimates the transplant effect by 1.549 on the log-hazard scale, and its 95% interval

never contains the true value. This bias is about the same size as the difference between the two models in the real data (-1.648 versus 0.020), so that difference is explained by immortal-time bias rather than by chance. The time-varying Cox model, which is correctly specified here, is nearly unbiased, with absolute bias at most 0.016 and coverage between 0.95 and 0.96.

TV-CSL does not improve on the time-varying Cox model at this sample size. Its estimates vary about twice as much across data sets (empirical standard deviation 0.52–0.71, compared with 0.27–0.34) and are biased by 0.15–0.19. Most of this bias arises in the 32% of data sets in which some training fold contains no prior-surgery deaths of one treatment status, which leaves the propensity poorly estimated in that fold; in these data sets, the average bias is 0.13–0.16, compared with 0.02–0.03 in the others. Information-based intervals cover the true value 89%–91% of the time, but sandwich intervals cover it only 60%–74% of the time, because the sandwich standard error is roughly half the actual variability. These results agree with the main simulation. When baseline risk is simple and the sample is small, estimating the additional nuisance functions costs precision without reducing bias, and information-based intervals are more reliable than sandwich intervals. We therefore use the time-varying Cox model as the primary analysis for the Stanford data.

Table 1: Summary Statistics for the Copilot Adoption Analysis. To reduce the risk of identifying the company, rank and office are reported in coarsened categories; offices are grouped into the eight largest offices and all remaining offices.

Characteristic	Category or summary	Overall
Seniority	Senior, count (%)	1,809 (54.4)
	Junior, count (%)	1,514 (45.6)
Rank (%)	Partner	20.0
	Associate	55.2
	Other	25.1
Office (%)	Eight largest offices	67.9
	Other offices	32.1
Working hours, first four observed weeks	Mean (SD); median [IQR]	39.0 (12.8); 40 [32, 46]
Missing working hours	Count (%)	—

Table 2: Cox Estimates That Ignore and Include AI Assistant Initiation Time

	Ignore treatment time		Time-varying treatment	
Effect	Hazard ratio (95% CI)	p-value	Hazard ratio (95% CI)	p-value
Used AI Assistant (senior)	2.11 (1.81, 2.47)	<0.001	1.63 (1.42, 1.88)	<0.001
Used AI Assistant (junior)	2.14 (1.59, 2.89)	<0.001	1.27 (1.08, 1.49)	0.003
Junior / senior ratio	1.01 (0.72, 1.42)	0.934	0.78 (0.64, 0.95)	0.015

Estimates are hazard ratios; values above 1 indicate a higher instantaneous rate of becoming a regular Copilot user. The junior/senior ratio and its confidence interval use the estimated covariance of the two group effects.

Table 3: Comparison of Time-Varying Cox and TV-CSL Estimates

	Time-varying	TV-CSL	
Effect	Cox	Marginal propensity	Time-varying propensity
	Hazard ratio (95% CI)	Hazard ratio (95% CI)	Hazard ratio (95% CI)
Used AI Assistant (senior)	1.63 (1.42, 1.88)	1.59 (1.35, 1.88)	1.51 (1.28, 1.79)
Used AI Assistant (junior)	1.27 (1.08, 1.49)	1.18 (0.98, 1.42)	1.23 (1.02, 1.48)
Junior / senior ratio	0.78 (0.64, 0.95)	0.74 (0.58, 0.94)	0.81 (0.63, 1.04)

Estimates are hazard ratios; values above 1 indicate a higher instantaneous rate of becoming a regular Copilot user. The junior/senior ratio and its confidence interval use the estimated covariance of the two group effects. TV-CSL (time-varying propensity) intervals use the cross-fitted sandwich variance estimator.

Table 4: MSE under linear η_0. All η_0 and propensity specifications. Lower is better.

Method	η_0 / prop. spec	$n = 200$	$n = 500$	$n = 1000$	$n = 2000$
S-Cox	complex	0.423	0.123	0.056	0.028
S-Cox	linear	0.219	0.079	0.033	0.018
S-Lasso	complex	0.297	0.105	0.049	0.025
S-Lasso	linear	0.219	0.079	0.033	0.018
TV-CSL	complex / intercept-only	1.495	0.695	0.458	0.259
TV-CSL	complex / marg. prop.	2.070	0.643	0.364	0.171
TV-CSL	complex / risk-set-adj.	0.808	0.485	0.379	0.273
TV-CSL	complex / time-var. oracle	3.121	0.742	0.296	0.109
TV-CSL	complex / time-var. prop.	2.457	0.511	0.225	0.095
TV-CSL	linear / intercept-only	0.378	0.113	0.042	0.023
TV-CSL	linear / marg. prop.	0.714	0.166	0.063	0.032
TV-CSL	linear / risk-set-adj.	0.351	0.091	0.036	0.019
TV-CSL	linear / time-var. oracle	1.059	0.209	0.081	0.039
TV-CSL	linear / time-var. prop.	0.917	0.162	0.064	0.030

Table 5: MSE under non-linear η_0. All η_0 and propensity specifications.

Method	η_0 / prop. spec	$n = 200$	$n = 500$	$n = 1000$	$n = 2000$
S-Cox	complex	0.382	0.168	0.114	0.094
S-Cox	linear	0.667	0.596	0.573	0.571
S-Lasso	complex	0.706	0.377	0.225	0.165
S-Lasso	linear	0.667	0.596	0.573	0.571
TV-CSL	complex / intercept-only	0.406	0.227	0.128	0.076
TV-CSL	complex / marg. prop.	0.923	0.235	0.095	0.040
TV-CSL	complex / risk-set-adj.	0.640	0.327	0.195	0.142
TV-CSL	complex / time-var. oracle	1.936	0.332	0.118	0.047
TV-CSL	complex / time-var. prop.	2.387	0.296	0.098	0.042
TV-CSL	linear / intercept-only	0.567	0.459	0.447	0.451
TV-CSL	linear / marg. prop.	0.761	0.197	0.104	0.080
TV-CSL	linear / risk-set-adj.	0.668	0.582	0.563	0.565
TV-CSL	linear / time-var. oracle	1.963	0.302	0.114	0.047
TV-CSL	linear / time-var. prop.	1.921	0.248	0.093	0.048

Table 6: Empirical 95% CI coverage under linear η_0 (linear η_0 spec). Averaged over $\beta_2, \beta_3, \beta_4$.

Method	prop. spec	$n = 200$	$n = 500$	$n = 1000$	$n = 2000$
S-Cox / S-Cox	—	0.933	0.943	0.955	0.942
S-Lasso / S-Cox	—	0.933	0.943	0.955	0.942
TV-CSL (naive)	intercept-only	0.909	0.896	0.908	0.878
TV-CSL (sandwich)	intercept-only	0.755	0.828	0.875	0.856
TV-CSL (naive)	marg. prop.	0.946	0.945	0.962	0.946
TV-CSL (sandwich)	marg. prop.	0.766	0.867	0.928	0.922
TV-CSL (naive)	risk-set-adj.	0.854	0.865	0.862	0.846
TV-CSL (sandwich)	risk-set-adj.	0.712	0.800	0.822	0.819
TV-CSL (naive)	time-var. oracle	0.916	0.940	0.956	0.944
TV-CSL (sandwich)	time-var. oracle	0.713	0.860	0.914	0.927
TV-CSL (naive)	time-var. prop.	0.924	0.943	0.955	0.946
TV-CSL (sandwich)	time-var. prop.	0.706	0.860	0.905	0.916

Table 7: Empirical 95% CI coverage under non-linear η_0 (complex η_0 spec). Averaged over $\beta_2, \beta_3, \beta_4$.

Method	prop. spec	$n = 200$	$n = 500$	$n = 1000$	$n = 2000$
S-Cox / S-Cox	—	0.914	0.839	0.736	0.545
S-Lasso / S-Cox	—	—	—	—	—
TV-CSL (naive)	intercept-only	0.709	0.515	0.458	0.395
TV-CSL (sandwich)	intercept-only	0.603	0.481	0.456	0.403
TV-CSL (naive)	marg. prop.	0.889	0.905	0.921	0.931
TV-CSL (sandwich)	marg. prop.	0.753	0.857	0.906	0.928
TV-CSL (naive)	risk-set-adj.	0.484	0.333	0.235	0.126
TV-CSL (sandwich)	risk-set-adj.	0.343	0.277	0.209	0.122
TV-CSL (naive)	time-var. oracle	0.845	0.900	0.924	0.945
TV-CSL (sandwich)	time-var. oracle	0.614	0.833	0.896	0.936
TV-CSL (naive)	time-var. prop.	0.857	0.920	0.929	0.933
TV-CSL (sandwich)	time-var. prop.	0.621	0.852	0.900	0.921

Table 8: Relative error of the complex outcome fit, $\{\text{RMSE of } \hat{\eta}_0\}/\operatorname{sd}(\eta_0)$, at $n = 4000$ and $\beta_0 = (0, 2, 2, 2)^\top$, averaged over two datasets.

w	0	0.25	0.5	0.75	1
Relative error	0.11	0.28	0.58	0.70	0.64

Table 9: TV-CSL under $\eta_0^{(0.25)}$ at $n = 4000$, by treatment-effect scale m. Bias is averaged over the three slope coefficients. Coverage entries are information/sandwich. Columns are marginal, time-varying, and oracle propensity.

	Bias			Coverage			MSE		
m	marg.	t.v.	oracle	marg.	t.v.	oracle	marg.	t.v.	oracle
0	−0.004	−0.006	−0.004	0.88/0.87	0.88/0.87	0.88/0.87	0.013	0.013	0.013
0.5	−0.007	−0.009	−0.003	0.93/0.91	0.91/0.90	0.93/0.92	0.013	0.014	0.013
1	−0.018	−0.012	0.001	0.91/0.89	0.91/0.89	0.93/0.91	0.015	0.016	0.015
1.5	−0.033	−0.022	0.000	0.89/0.88	0.91/0.90	0.93/0.93	0.018	0.019	0.018
2	−0.048	−0.031	−0.002	0.83/0.80	0.90/0.87	0.91/0.91	0.025	0.024	0.022

Table 10: Sensitivity of AI Assistant Effect Estimates to Cohort and Tie Convention

	New users ($n = 2,026$)		All lawyers ($n = 3,322$)	
Effect	$W(t) = 1(A < t)^{\dagger}$	$W(t) = 1(A \leq t)$	$W(t) = 1(A < t)$	$W(t) = 1(A \leq t)$
Cox, ignoring treatment time				
Used AI Assistant (senior)	0.748 (0.591, 0.904)	0.748 (0.591, 0.904)	0.761 (0.616, 0.907)	0.761 (0.616, 0.907)
Used AI Assistant (junior)	0.762 (0.461, 1.062)	0.762 (0.461, 1.062)	0.694 (0.413, 0.975)	0.694 (0.413, 0.975)
Junior − senior difference	0.014 (-0.322, 0.350)	0.014 (-0.322, 0.350)	-0.067 (-0.380, 0.245)	-0.067 (-0.380, 0.245)
Time-varying Cox				
Used AI Assistant (senior)	0.489 (0.348, 0.629)	0.796 (0.660, 0.931)	0.526 (0.416, 0.636)	0.814 (0.704, 0.923)
Used AI Assistant (junior)	0.239 (0.081, 0.397)	0.703 (0.540, 0.867)	0.136 (0.015, 0.257)	0.560 (0.429, 0.691)
Junior − senior difference	-0.250 (-0.452, -0.048)	-0.092 (-0.294, 0.110)	-0.390 (-0.540, -0.239)	-0.254 (-0.412, -0.096)
TV-CSL, marginal propensity				
Used AI Assistant (senior)	0.466 (0.301, 0.631)	0.718 (0.546, 0.890)	0.507 (0.383, 0.631)	0.764 (0.636, 0.892)
Used AI Assistant (junior)	0.165 (-0.019, 0.349)	0.489 (0.303, 0.675)	0.037 (-0.101, 0.174)	0.423 (0.274, 0.571)
Junior − senior difference	-0.301 (-0.542, -0.060)	-0.229 (-0.476, 0.018)	-0.470 (-0.652, -0.289)	-0.342 (-0.535, -0.149)
TV-CSL, time-varying propensity				
Used AI Assistant (senior)	0.415 (0.247, 0.582)	0.658 (0.482, 0.833)	0.439 (0.312, 0.566)	0.693 (0.561, 0.824)
Used AI Assistant (junior)	0.205 (0.020, 0.390)	0.576 (0.383, 0.768)	0.046 (-0.095, 0.187)	0.449 (0.292, 0.605)
Junior − senior difference	-0.210 (-0.455, 0.036)	-0.082 (-0.338, 0.173)	-0.394 (-0.582, -0.205)	-0.244 (-0.448, -0.041)

Estimates (95% CI) on the log-hazard-ratio scale. †Primary analysis. New users have no recorded AI Assistant use before time zero; the all-lawyers cohort adds prior users and adjusts for prior use. Both cohorts exclude lawyers who became regular users in their entry week.

Table 11: Summary Statistics for the Stanford Heart Transplant Data

Variable	Mean	SD
Age	45.17	9.80
Prior surgery	0.16	0.36
Year of acceptance	3.36	1.86
Received transplant	0.67	0.47

Table 12: Cox Estimates That Ignore and Include Transplant Time

	Ignore transplant time		Include transplant time	
Effect	Estimate (95% CI)	p-value	Estimate (95% CI)	p-value
Transplant	-1.648 (-2.207, -1.089)	<0.001	0.020 (-0.616, 0.657)	0.951
Age × transplant	-0.228 (-0.782, 0.326)	0.420	0.296 (-0.223, 0.816)	0.263
Year × transplant	0.143 (-0.360, 0.645)	0.578	0.394 (-0.116, 0.904)	0.130

Estimates are log hazard ratios for death; negative values indicate lower mortality after transplant. Age and year of acceptance are standardized. All models adjust for age, year of acceptance and prior surgery; prior surgery is not an effect modifier.

Table 13: Semi-Synthetic Bias and Coverage at the Stanford Sample Size

	Transplant (0.02)		Age × transplant (0.30)		Year × transplant (0.39)	
Method	Bias	Coverage	Bias	Coverage	Bias	Coverage
Cox, ignoring transplant time	-1.549	0.00	-0.235	0.81	-0.099	0.87
Time-varying Cox	-0.016	0.96	0.009	0.96	0.008	0.95
TV-CSL, marginal propensity	-0.171	0.91/0.74	0.159	0.90/0.64	0.193	0.90/0.71
TV-CSL, time-varying propensity	-0.155	0.91/0.72	0.167	0.91/0.60	0.187	0.89/0.71

Results over 500 replicates of $n = 103$. True values (in parentheses) are the time-varying Cox estimates from the real data. Covariates and censoring dates are the real ones; transplant and death times are redrawn from Cox models fitted to the real data. Coverage is of nominal 95% intervals; TV-CSL entries are information/sandwich.